



**The Abdus Salam
International Centre for Theoretical Physics
Strada Costiera, 11 I - 34151 Trieste, Italy
Earth System Physics Section - ESP**

**Regional Climatic Model RegCM
User's Guide
Version 4.6**

Trieste, Italy February 19, 2022

Filippo Giorgi, Fabien Solmon, Graziano Giuliani

Contents

1	Release Notes	5
2	Obtaining the model	7
2.1	Simple Model User	7
2.2	Model Developer	7
3	Installation procedure	8
3.1	Software requirements	8
3.2	Configuring build	9
3.2.1	Model configuration at build stage	10
3.3	Build the model executables	10
4	Accessing global datasets	11
4.1	Global dataset directory Layout	11
4.2	Static Surface Dataset	11
4.3	CLM Dataset	12
4.4	CLM 4.5 Dataset	12
4.5	Sea Surface Temperature	13
4.6	Atmosphere and Land temperature Global Dataset	13
5	Running a test simulation using the model	14
5.1	Setting up the run environment	14
5.2	Create the DOMAIN file using terrain	15
5.3	Create the SST using the sst program	16
5.4	Create the ICBC files using the icbc program	16
5.5	First RegCM model simulation	17
6	Localizing the model and running your simulation	19
6.1	The commented namelist	19
6.1.1	coreparam namelist	19
6.1.2	dimparam namelist	20
6.1.3	geoparam namelist	20
6.1.4	terrainparam namelist	22
6.1.5	globdatparam namelist	23
6.1.6	fnestparam namelist	24
6.1.7	restartparam namelist	24
6.1.8	timeparam namelist	25
6.1.9	outparam namelist	25
6.1.10	physicsparam namelist	27
6.1.11	dynparam namelist	28
6.1.12	referenceatm and nonhydroparam namelist	29
6.1.13	hydroparam namelist	29

6.1.14	boundaryparam namelist	29
6.1.15	cldparam namelist	30
6.1.16	subexparam namelist	31
6.1.17	microparam namelist	31
6.1.18	grellparam, emanparam, tiedtkeparam and kfparam namelists	32
6.1.19	holtslagparam namelist	33
6.1.20	uwparam namelist	34
6.1.21	slabocparam namelist	34
6.1.22	perturbparam namelist	34
6.1.23	tweakparam namelist	35
6.1.24	rrtmparam namelist	35
6.1.25	chemparam namelist	35
6.1.26	debugparam namelist	39
6.2	The CLM options	39
6.3	The CLM 4.5 options	40
6.4	Sensitivity experiments hint	41
7	How To	42
7.1	Introduction	42
7.1.1	1D Lake Model	42
8	Postprocessing tools	43
8.1	Command line tools	43
8.1.1	netCDF library tools	43
8.1.2	NetCDF operators NCO	44
8.1.3	Climate data Operators CDO	45
8.2	GrADS program	45
8.2.1	GrADS limits	45
8.3	CISL's NCL : NCAR Command Language	46
8.4	R Statistical Computing Language	46
8.5	Non free tools	46
9	Getting help and reporting bugs	47
9.1	The Gforge site	47
10	Appendices	49
10.1	Identify Processor	49
10.2	Chose compiler	50
10.3	Environment setup	51
10.4	Pre requisite library installation	51

Chapter 1

Release Notes

The Regional Climate Model developed at ICTP has now reached its 4.5 release. The code base is actively developed by a community of developers internal and external to ICTP, and this work is merged on the Gforge site on gforge.ictp.it site.

The main new technical features of the release 4.5 are summarized in the following points:

- The MM5 non-hydrostatic dynamical core has been ported to the ICTP RegCM without removing the existing hydrostatic core. Now the user can select in the input namelist which of the two dynamical cores he wish to use.
- The SAV file strategy has changed to allow a better integration of the RegCM into the RegESM framework and the user is required to modify the setting from previous version namelist files.
- The source for the topography is now the GMTED dataset which is an update to the USGS GTOPO dataset.
- The DUST 12 bin option has been added to the aerosol simulation types
- The chemistry/aerosol option now works along the CLM4.5 option
- The model supports soil moisture initialization from a previous run for both the BATS and the CLM4.5 surface models.

Bug Fixing:

- Bugs present in the CN schemes for the CLM4.5 implemented in RegCM have been fixed
- A problematic interaction between the UW pbl and the new Microphysics has been fixed

Next release V 5.X :

- New dynamical core from Giovanni Tumolo semi-implicit, semi-Lagrangian, p-adaptive Discontinuous Galerkin method three dimensional model.

The model code is in Fortran 2003 ANSI standard. The development is done on Linux boxes, and the model is known to run on IBM AIXTMplatforms, MacOSTMplatforms. No porting effort has been done towards non Unix-like Operating Systems. We will for this User Guide assume that the reference platform is a recent Linux distributon with a `bash` shell. Typographical convention is the following:

Table 1.1: Conventions

<code>\$></code>	normal shell prompt
<code>#></code>	root shell prompt
<code>\$SHELL_VARIABLE</code>	a shell variable

Any shell variable is supposed to be set by the User with the following example syntax:

```
$> export REGCM_ROOT="/home/user/RegCM-4.5.0"
```

Hope you will find this document useful. Any error found belongs to the RegCNET and can be reported to be corrected in future revisions. Enjoy.

Chapter 2

Obtaining the model

2.1 Simple Model User

A packed archive file with the model code can be downloaded from:

`http://gforge.ictp.it/gf/project/regcm/frs`

and it can be later on decompressed and unpacked using:

```
$> tar -zxvf RegCM-4.5.0.tar.gz
```

2.2 Model Developer

If you plan to become a model developer, source code can be obtained via svn. The RegCM team strongly encourages the contributing developers to enroll on the gforge site to always be up to date and to check on-line all the news of the package.

<https://gforge.ictp.it/gf/project/regcm>

The correct procedure is first to register on the G-forge site, then ask the ICTP scientific team head Filippo Giorgi to be enrolled as a model developer. After being officially granted the status, you will gain access to the model subversion repository.

Check that **Subversion** software is installed on your machine typing the following command:

```
$> svn --version
```

If your system answers `command not found`, refer to your System Administrator or software installation manual of your OS to install the subversion software. As an example, on Scientific Linux the command to install it as root is:

```
#> yum install subversion
```

If Subversion is installed, after enrolled just type the following command:

```
$> svn checkout https://gforge.ictp.it/svn/regcm/branches/regcm-core
```

To setup the model code from the SVN for your system, run the `bootstrap.sh` script:

```
$> ./bootstrap.sh
```

The system must have installed the `autoconf`, `automake` and `libtool` programs.

Chapter 3

Installation procedure

Whatever method is chosen to download the code, we assume that you have now on your working directory a new directory, named `RegCM-4.5.x`. That directory will be for the rest of this guide referred as `$REGCM_ROOT`.

All the operations to build the model binaries will be performed in this directory.

3.1 Software requirements

In order to configure and install the RegCM code, the following software are needed:

1. GNU Make program
2. Fortran 2003 compiler
3. One in:
 - (a) C compiler for the serial option (`enable-mpi-serial` at configure)
 - (b) MPI2 Message Passing Library compiled with the above fortran compiler for parallel runs using multiple core single machines or cluster of machines (default). Source code for the implementation code was tested with can be obtained at:
<http://www.open-mpi.org>
4. netCDF (*Rew and Davis* (1990)) Format I/O library compiled with the above Fortran compiler with netCDF4 format support. Source code can be found from
<ftp://ftp.unidata.ucar.edu/pub/netcdf>

Optional requirements strongly suggested are :

1. NCO netCDF Operators for mangning netCDF file. Most Linux distribution have this already packed, and you should refer to your System Administrator or OS Software Installation manual to obtain it. Source code is at:
<http://nco.sourceforge.net/src>
2. CDO Climatic data Operators for managing netCDF file. Most Linux distribution have this already packed, and you should refer to your System Administrator or OS Software Installation manual to obtain it. Source code is at:
<https://code.zmaw.de/projects/cdo/files>
3. A Scientific Plotting and Data Analysis Software such as:
 - IGES GrADS 2.0 Graphical Analysis and Display System. Convenient helpers are packed in RegCM to use GrADS with RegCM netCDF output files. Binaries and source code can be obtained from
<http://www.iges.org/grads/downloads.html>

- NCL, NCAR CISL Command Language. The NCL can read netCDF output files, and sample scripts can be found in the *Tools/Scripts/NCL* directory. Binaries and source code can be obtained from <http://www.ncl.ucar.edu>

4. A quick viewer for netCDF files like NcView:
http://meteora.ucsd.edu/pierce/ncview_home_page.html

A script is shipped in the RegCM codebase in the Tools directory to help user compile and install required packages, and its usage is detailed in chapter 10.

3.2 Configuring build

The RegCM Version 4.5 is configured by a configure script, which will select the known working configuration for the supported architectures.

Currently tested and supported configurations (OS/Compiler) are:

1. Linux with GNU gfortran compiler version ≥ 4.8
2. Linux with IntelTMifort compiler version ≥ 13.0
3. Linux with PortlandTMpgf95 compiler version ≥ 12.0
4. Mac OSTMwith gfortran compiler ≥ 4.6 from MacPorts
5. IBM AIXTMwith xlf2003 compiler

The 4.5 version of the RegCM model relies on the standard GNU autotools to configure and build the model code.

The first step is to change working directory to \$REGCM_ROOT and run the configure script giving as arguments the chosen compilers:

```
$> cd $REGCM_ROOT
$> ./configure # or, for intel : ./configure CC=icc FC=ifort
```

To know the list of arguments that can be given to the configure script, the script can be launched with the --help command line argument.

```
$> ./configure --help
```

The useful arguments to successfully build the model are:

```
--with-netcdf      Path to NetCDF installation (default: NETCDF
                   environment)
--with-hdf5        Path to HDF5 installation (default: HDF5
                   environment)
--with-szip        Path to SZIP installation (default: SZIP
                   environment)
CC=                C compiler command
CFLAGS=            C compiler flags
LDFLAGS=           linker flags, e.g. -L<lib dir> if you have libraries in a
                   nonstandard directory <lib dir>
LIBS=              libraries to pass to the linker, e.g. -l<library>
CPPFLAGS=          (Objective) C/C++ preprocessor flags, e.g. -I<include dir> if
                   you have headers in a nonstandard directory <include dir>
CPP=               C preprocessor
FC=                Fortran compiler command
FCFLAGS=           Fortran compiler flags
MPIFC=             MPI Fortran compiler command
```

3.2.1 Model configuration at build stage

1. Enable debug

`--enable-debug` Enable debugging flags and per processor log file

If enabled, the model will be compiled using debug flags for the compiler, which will allow the use of a debugger such as gdb. More diagnostics will also be generated during model run. The default is to build production binaries with all optimization flags turned on.

2. Serial code using stub MPI library

`--enable-mpiserial` Use the included MPI replacement library for single processor

The model is coded to use an MPI2 library to run in parallel mode using multiple cores/processors or run on a cluster. To enable instead a serial compilation option, a stub MPI library with empty callbacks needs to be compiled and linked to the executable. The RegCM team strongly suggest to build MPI enabled model also on standalone systems, to take advantage of the multicore capabilities of any modern processor.

3. CLM option

`--enable-clm` Supply this option if you plan on using CLM option.

This option switches off the default Land model of RegCM (derived from BATS1e), and enables the use of the Community Land Model V3.5 inside RegCM. The default is to use the RegCM BATS Land Model.

4. CLM 4.5 option

`--enable-clm45` Supply this option if you plan on using CLM45 option.

This option switches off the default Land model of RegCM (derived from BATS1e), and enables the use of the Community Land Model V4.5 inside RegCM. The default is to use the RegCM BATS Land Model.

3.3 Build the model executables

Now that everything is hopefully configured, you may use the `make` program to build executables.

```
$> make
```

This target will builds all model parts. The compilation is started in the whole model tree (PreProc, Main and PostProc). Lot of messages will appear on screen, abd at the end all executables are built int the source directories. To copy them to a `bin` directory in the model root or into a `bin` directory in the path specified with the prefix argument to the `configure` script, esplicitly issue the command:

```
$> make install
```

Congratulations! You can now go to next step and run a test simulation.

Chapter 4

Accessing global datasets

The first step to run a test simulation is to obtain static data to localize model DOMAIN and Atmosphere and Ocean global model dataset to build initial and boundary conditions ICBC to run a local area simulation.

ICTP maintains a public accessible web repository of datasets on:

<http://clima-dods.ictp.it/regcm4>

We will in the following substitute this URL with a shell variable:

```
$> export ICTP_DATASITE=http://clima-dods.ictp.it/regcm4
```

As of now you are requested to download required global data on your local disk storage before any run attempt. In the future, the ICTP ESP team has planned to make available an OpenDAP THREDDS Server to give remote access to global dataset for creating DOMAIN and ICBC without the need to download the global dataset, but just the required subset in space and time, using the ICTP web server capabilities to create that subset.

Our advice to you is to use handy transfer program such as uget, but below we will show you how to use command line download tools curl and wget to get data.

4.1 Global dataset directory Layout

You are suggested to establish a convenient location for global datasets on your local storage. Keep in mind that required space for a year of global data can be as large as 8 GBytes.

Having this in mind, we will now consider that you the user have identified on your system or have network access to such a storage resource to store say 100 GB of data, and have it reachable on your system under the \$REGCM_GLOBEDAT location. On this directory, you are required to make the following directories:

```
$> cd $REGCM_GLOBEDAT
$> mkdir SURFACE CLM CLM45 SST EIN15
```

This does not fill all possible global data sources paths, but will be enough for the scope of running the model for testing its capabilities.

4.2 Static Surface Dataset

The model needs to be localized on a particular DOMAIN. The needed information are topography, land type classification and optionally lake depth (to run the Hostetler lake model) and soil texture classification (to run the chemistry option with DUST enabled).

This means downloading four files, which are global archives at 30second horizontal resolution on a global latitude-longitude grid of the above data.

```
$> cd $REGCM_GLOBEDAT
$> cd SURFACE
$> curl -o GTOPO_DEM_30s.nc ${ICTP_DATASITE}/SURFACE/GTOPO_DEM_30s.nc
$> curl -o GLCC_BATS_30s.nc ${ICTP_DATASITE}/SURFACE/GLCC_BATS_30s.nc
$> curl -o GLZB_SOIL_30s.nc ${ICTP_DATASITE}/SURFACE/GLZB_SOIL_30s.nc
```

Optional Lake dataset:

```
$> cd $REGCM_GLOBEDAT
$> cd SURFACE
$> curl -o ETOPO_BT_30s.nc ${ICTP_DATASITE}/SURFACE/ETOPO_BT_30s.nc
```

4.3 CLM Dataset

If you are planning to enable the CLM option in the model, you will need a series of files with global land surface characteristics datasets.

```
$> cd $REGCM_GLOBEDAT
$> cd CLM
$> curl -o mksrf_fmax.nc ${ICTP_DATASITE}/CLM/mksrf_fmax.nc
$> curl -o mksrf_glacier.nc ${ICTP_DATASITE}/CLM/mksrf_glacier.nc
$> curl -o mksrf_lai.nc ${ICTP_DATASITE}/CLM/mksrf_lai.nc
$> curl -o mksrf_lanwat.nc ${ICTP_DATASITE}/CLM/mksrf_lanwat.nc
$> curl -o mksrf_navyoro_20min.nc ${ICTP_DATASITE}/CLM/mksrf_navyoro_20min.nc
$> curl -o mksrf_pft.nc ${ICTP_DATASITE}/CLM/mksrf_pft.nc
$> curl -o mksrf_soicol_clm2.nc ${ICTP_DATASITE}/CLM/mksrf_soicol_clm2.nc
$> curl -o mksrf_soitex.10level.nc ${ICTP_DATASITE}/CLM/mksrf_soitex.10level.nc
$> curl -o mksrf_urban.nc ${ICTP_DATASITE}/CLM/mksrf_urban.nc
$> curl -o pft-physiology.c070207 ${ICTP_DATASITE}/CLM/pft-physiology.c070207
$> curl -o pft-physiology.c070207.readme \
> ${ICTP_DATASITE}/CLM/pft-physiology.c070207.readme
$> curl -o rdir.05.061026 ${ICTP_DATASITE}/CLM/rdir.05.061026
```

This is the input file for the `clm2rcm` program (see at 6.2).

4.4 CLM 4.5 Dataset

If you are planning to enable the CLM 4.5 option in the model, you will need a series of files with global land surface characteristics datasets.

```
$> cd $REGCM_GLOBEDAT
$> cd CLM45
$> mkdir megan pftdata snicardata surface
$> for dir in megan pftdata snicardata surface; do cd $dir; \
    wget ${ICTP_DATASITE}/CLM45/$dir/ -O - | \
    wget -A ".nc" -ll --no-parent --base=${ICTP_DATASITE}/CLM45/$dir/ \
    -nd -Fri -; done
```

This is the input file for the `mksurfddata` program (see at 6.3).

4.5 Sea Surface Temperature

The model needs a global SST dataset to feed the model with ocean temperature. You have multiple choices for SST data, but we will for now for our test run download just CAC OISST weekly for the period 1981 - present.

```
$> cd $REGCM_GLOBEDAT
$> cd SST
$> CDCSITE="ftp.cdc.noaa.gov/pub/Datasets/noaa.oisst.v2"
$> curl -o sst.wkmean.1981-1989.nc \
> ftp://$CDCSITE/sst.wkmean.1981-1989.nc
$> curl -o sst.wkmean.1990-present.nc \
> ftp://$CDCSITE/sst.wkmean.1990-present.nc
```

4.6 Atmosphere and Land temperature Global Dataset

The model needs to build initial and boundary conditions for the regional scale, interpolating on the RegCM grid the data from a Global Climatic Model output. The GCM dataset can come from any of the supported models, but we will for now for our test run download just the EIN15 dataset for the year 1990 (Jan 01 00:00:00 UTC to Dec 31 18:00:00 UTC)

```
$> cd $REGCM_GLOBEDAT
$> cd EIN15
$> mkdir 1990
$> cd 1990
$> for type in "air hgt rhum uwnd vwnd"
> do
>   for hh in "00 06 12 18"
>   do
>     curl -o ${type}.1990.${hh}.nc \
>         ${ICTP_DATASITE}/EIN15/1990/${type}.1990.${hh}.nc
>   done
> done
```

With this datasets we are now ready to go through the RegCM Little Tutorial in the next chapter of this User Guide.

Chapter 5

Running a test simulation using the model

We will in this chapter go through a sample session in using the model with a sample configuration file prepared for this task.

5.1 Setting up the run environment

The model executables prepared in chapter 3 are waiting for us to use them. So let's give them a chance.

The model test run proposed here requires around 100Mb of disk space to store the DOMAIN and ICBC in input and the output files. We will assume here that you, the user, have already established a convenient directory on a disk partition with enough space identified in the following discussion with `$REGCM_RUN`

We will setup in this directory a standard environment where the model can be executed for the purpose of learning how to use it.

```
$> cd $REGCM_RUN
$> mkdir input output
$> ln -sf $REGCM_ROOT/bin .
$> cp $REGCM_ROOT/Testing/test_001.in .
$> cd $REGCM_RUN
```

Now we are ready to modify the input namelist file to reflect this directory layout. A namelist file in FORTRAN is a convenient way to give input to a program in a formatted file, read at runtime by the program to setup its execution behaviour. So the next step is somewhat tricky, as you need to edit the namelist file respecting its well defined syntax. Open your preferred text file editor and load the `test_001.in` file. You will need to modify for the scope of the present tutorial the following lines:

```
FROM:
  dirter = '/set/this/to/where/your/domain/file/is',
TO:
  dirter = 'input/',

FROM:
  inpter = '/set/this/to/where/your/surface/dataset/is',
TO:
  inpter = '$REGCM_GLOBEDAT',
```

where `$REGCM_GLOBEDAT` is the directory where input data have been downloaded in chapter 4.

```
FROM:
  dirglob = '/set/this/to/where/your/icbc/for/model/is',
TO:
  dirglob = 'input/',
```

```
FROM:
  inpglob = '/set/this/to/where/your/input/global/data/is',
TO:
  inpglob = '$REGCM_GLOBEDAT',
```

and last bits:

```
FROM:
  dirout='/set/this/to/where/your/output/files/will/be/written'
TO:
  dirout='output/'
```

These modifications just reflect the above directory layout proposed for this tutorial, and any of these paths can point anywhere on your system disks. The path is limited to 256 characters. We are now ready to execute the first program of the RegCM model.

5.2 Create the DOMAIN file using terrain

The first step is to create the DOMAIN file to localize the model on a world region. The program which does this for you reading the global databases is `terrain`.

To launch the `terrain` program, enter the following commands:

```
$> cd $REGCM_RUN
$> ./bin/terrain test_001.in
```

If everything is correctly configured up to this point, the model should print something on stdout, and the last lines will be:

```
Grid data written to output file
Successfully completed terrain fields generation
```

In the input directory the program will write the following two files:

```
$> ls input
test_001_DOMAIN000.nc test_001_LANDUSE
```

The DOMAIN file contains the localized topography and landuse databases, as well as projection information and land sea mask. The second file is an ASCII encoded version of the landuse, used for modifying it on request. We will cover its usage later on. To have a quick look at the DOMAIN file content, you may want to use the `GrADSNcPlot` program:

```
$> ./bin/GrADSNcPlot input/test_001_DOMAIN000.nc
```

If not familiar with GrADS program, enter in sequence the following commands at the `ga->` prompt:

```
ga-> q file
ga-> set gxout shaded
ga-> set mpdset hires
ga-> set cint 50
ga-> d topo
ga-> c
ga-> set cint 1
ga-> d landuse
ga-> quit
```

this will plot the topography and the landuse on the X11 window.

5.3 Create the SST using the sst program

We are now ready to create the Sea Surface Temperature for the model, reading a global dataset. The program which does this for you is the `sst` program, which is executed with the following commands:

```
$> cd $REGCM_RUN
$> ./bin/sst test_001.in
```

If everything is correctly configured up to this point, the model should print something on stdout, and the last line will be:

```
Successfully generated SST
```

The input directory now contains a new file:

```
$> ls input
test_001_DOMAIN000.nc test_001_LANDUSE test_001_SST.nc
```

The SST file contains the Sea Surface temperature to be used in generating the Initial and Boundary Conditions for the model for the period specified in the namelist file. Again you may want to use the GrADSNCPlot program to look at file content:

```
$> ./bin/GrADSNCPlot input/test_001_SST.nc
```

If not familiar with GrADS program, enter in sequence the following commands at the `ga->` prompt:

```
ga-> q file
ga-> set gxout shaded
ga-> set mpdset hires
ga-> set cint 2
ga-> d sst
ga-> quit
```

this will plot the interpolated sst field on the X11 window.

5.4 Create the ICBC files using the icbc program

Next step is to create the ICBC (Initial Condition, Boundary Conditions) for the model itself. The program which does this for you is the `icbc` program, executed with the following commands:

```
$> cd $REGCM_RUN
$> ./bin/icbc test_001.in
```

If everything is correctly configured up to this point, the model should print something on stdout, and the last line will be:

```
Successfully completed ICBC
```

The input directory now contains two more files:

```
$> ls -l input
test_001_DOMAIN000.nc
test_001_ICBC.1990060100.nc
test_001_ICBC.1990070100.nc
test_001_LANDUSE
test_001_SST.nc
```


The ICBC files contain the surface pressure, surface temperature, horizontal 3D wind components, 3D temperature and mixing ratio for the RegCM domain for the period and time resolution specified in the input file. Again you may want to use the GrADSNcPlot program to look at file content:

```
$> ./bin/GrADSNcPlot input/test_001_ICBC.1990060100.nc
```

If not familiar with GrADS program, enter in sequence the following commands at the ga-> prompt:

```
ga-> q file
ga-> set gxout shaded
ga-> set mpdset hires
ga-> set cint 2
ga-> d ts
ga-> c
ga-> set lon 10
ga-> set lat 43
ga-> set t 1 last
ga-> d ts
ga-> quit
```

this will plot the interpolated surface temperature field on the X11 window, first at first time step and then a time section in one of the domain points for a whole month.

We are now ready to run the model!

5.5 First RegCM model simulation

The model has now all needed data to allow you to launch a test simulation, the final goal of our little tutorial.

The model command line now will differ if you have prepared the Serial or the MPI version. For the MPI enabled version we will assume that your machine is a dual core processor (baseline for current machines, even for laptops). Change the `-np 2` argument to the number of processors you have on Your platform (on my laptop QuadCore I use `-np 4`).

- MPI version ¹

```
$> cd $REGCM_RUN
$> mpirun -np 8 ./bin/regcmMPI test_001.in
```

- Serial version ²

```
$> cd $REGCM_RUN
$> ./bin/regcmSerial test_001.in
```

Now the model will start running, and a series of diagnostic messages will be printed on screen. As this is a simulation known to behave well, no stoppers will appear, so you may want now to have a coffee break and come back in 10 minutes from now.

At the end of the run, the model will print the following message:

```
RegCM V4 simulation successfully reached end
```

The output directory now contains four files:

```
$> ls output
test_001_ATM.1990060100.nc test_001_SRF.1990060100.nc
test_001_RAD.1990060100.nc test_001_SAV.1990070100.nc
```

¹Use `regcmMPICLM` if the CLM version has been configured

²Deprecated. Support will be dropped in future releases.

the ATM file contains the atmosphere status from the model, the SRF file contains the surface diagnostic variables, and the RAD file contains radiation fluxes information. The SAV file stores the status of the model at the end of the simulation period to enable a restart, thus allowing a long simulation period to be splitted in shorter simulations.

To have a look for example at surface fields, you may want to use the following command:

```
$> ./bin/GrADSNcPlot output/test_001_SRF.1990060100.nc
```

Assuming the previous crash course in using GrADS was received, you should be able to plot the variables in the file.

This is the end of this little tutorial, and in the next chapter we will examine how to configure the model for your research needs.

Chapter 6

Localizing the model and running your simulation

We will examine in this chapter in more detail the namelist configuration file, to give you the User a deeper knowledge of model capabilities.

6.1 The commented namelist

In this section we will show you the commented namelist input file you will find under `$REGCM_ROOT/Doc` with the name `README.namelist`. All model programs seen so far, with the exception of the GrADS helper program, use as input this namelist file, which is unique to a particular simulation. The model input namelist file is composed by a number of different namelists, each one devoted to configuring the model capabilities. A namelist in the namelist file is identified with a starting `&` character followed by namelist name, and ends on a single line with the `/` character.

6.1.1 coreparam namelist

This namelist controls which dynamical core is used in the model.

```
&coreparam
idynamic = 1, ! Choice of dynamical core
              ! 1 = MM4 hydrostatic core
              ! 2 = MM5 NON hydrostatic core
/
```

1. For details about the two dynamical cores, refer to the Reference Manual.
2. The hydrostatic core is cheaper computationally, but physically it should be limited to resolution greater than *15km*. For higher resolution, the non hydrostatic core should be used.
3. The output variables of the two dynamical cores differ in the ATM files because of the different state variables used. Vertical levels also differ.
4. The ICBC for the non-hydrostatic cannot be used for the hydrostatic run and the other way round.
5. Nesting using the `FNEST` option is supported in these configurations:
 - (a) hydrostatic nested into hydrostatic
 - (b) non-hydrostatic nested into hydrostatic
 - (c) non-hydrostatic nested into non-hydrostatic
6. Selecting the non-hydrostatic core, the `nonhydroparam` namelist is read in.

6.1.2 dimparam namelist

This namelist contains the base X,Y,Z domain dimension information, used by the model dynamic memory allocator to request the Operating System the memory space to store the model internal variables.

```
&dimparam
iy      = 34,    ! This is number of points in the N/S direction
jx      = 48,    ! This is number of points in the E/W direction
kz      = 23,    ! Number of vertical levels
dsmin   = 0.01,  ! Minimum sigma spacing (only used if kz is not 14, 18, 23, 41)
dsmax   = 0.05,  ! Maximum sigma spacing (only used if kz is not 14, 18, 23, 41)
nsg      = 1,    ! For subgridding, number of points to decompose. If nsg=1,
                  ! no subgridding is performed. CLM3.5 does NOT work with
                  ! subgridding enabled.
njxcpus = -1,    ! Number of CPUS to be used in the jx (lon) dimension.
                  ! If <=0 , the executable will try to figure out a suitable
                  ! decomposition.
niycpus = -1,    ! Number of CPUS to be used in the iy (lat) dimension.
                  ! If <=0 , the executable will try to figure out a suitable
                  ! decomposition.
/
```

The things you need to know here:

1. In the current version the model parallelizes execution dividing the work between the processors, with the minimum work per processor is 9 points or a box 3×3 , so the maximum theoretical number of processors which can be used in a parallel run for the above configuration is limited by the domain the overall number of horizontal grid points.
2. If a custom number of sigma level is chosen (not 14, 18, 23 or 41), the actual sigma values are calculated minimizing the a, b coefficients for the equation:

$$d\sigma(i) = \max(dsmax * a^{i-1} * b^{0.5*(i-2)*(i-1)}, dsmin) \quad (6.1)$$

derived from the recursive relation:

$$d\sigma(i) = a(i) * d\sigma(i-1) \quad (6.2)$$

where $a(i) = b * a(i-1)$. We at ICTP normally use 23 levels for the non-hydrostatic core and 18 for the hydrostatic core at 50 km.

3. Specifying an *nsg* number greater than one triggers the subgrid BATS/CLM45 model on. There is no plan to extend this feature to CLM3.5 model. This affects only surface variable calculations. All dynamical variables are calculated still on the coarser grid. Rain in the current implementation is also calculated on the coarser grid.
4. The *njxcpus*, *niycpus* parameters can be used to force a particular domain decomposition for any particular hardware architecture. Normally the algorithm in the model code should be able to fix the supposedly optimally balanced decomposition.

6.1.3 geoparam namelist

This namelist is used by the `terrain` program to geolocate the model grid on the earth surface. The RegCM model uses a limited number of projection engines. The value here are used by the other model programs to assert consistency with the geolocation information written by the `terrain` program in the `DOMAIN` file.

The first step in any application is the selection of model domain and resolution. There are no strict rules for this selection, which in fact is mostly determined by the nature of the problem and the availability of computing resources. The domain should be large enough to allow the model to develop its own circulations and to include

all relevant forcings and processes, and the resolution should be high enough to capture local processes of interest (e.g. due to complex topography or land surface).

On the other hand the model computational cost increases rapidly with resolution and domain size, so a compromise needs to be usually reached between all these factors.

This is usually achieved by experience, understanding of the problem or trial and error, however one tip to remember is to avoid that the boundaries of the domain cross major topographical systems.

This is because the mismatch in the resolution of the coarse scale lateral driving fields and the model fields in the presence of steep topography may generate spurious local effects (e.g. localized precipitation areas) which can affect the model behavior, at least in adjacent areas.

```
&geoparam
iproj = 'LAMCON', ! Domain cartographic projection. Supported values are:
                ! 'LAMCON', Lambert conformal.
                ! 'POLSTR', Polar stereographic.
                ! 'NORMER', Normal Mercator.
                ! 'ROTMER', Rotated Mercator.
ds = 60.0,      ! Grid point horizontal resolution in km
ptop = 5.0,     ! Pressure of model top in cbar
clat = 45.39,   ! Central latitude of model domain in degrees
                ! North hemisphere is positive
clon = 13.48,   ! Central longitude of model domain in degrees
                ! West is negative.
cntri = -1      ! S/N index of projection center in domain.
                ! Defaults to IY/2.0
cntrj = -1      ! W/E index of projection center in domain.
                ! Defaults to JX/2.0
plat = 45.39,   ! Pole latitude (only for rotated Mercator Proj)
plon = 13.48,   ! Pole longitude (only for rotated Mercator Proj)
truelatl = 30.0, ! Lambert true latitude (low latitude side)
truelath = 60.0, ! Lambert true latitude (high latitude side)
i_band = 0,     ! Use this to enable a tropical band. In this case the ds,
                ! iproj, clat, clon parameters are not considered.
i_crm = 0,     ! Use this to enable doubly-periodic domains for idealized
                ! simulations. Forces i_band = 1.
/
```

The things you need to know here:

1. The different projection engines produce better results depending on the position and extent of the domain. In particular, regardless of hemisphere:
 - Middle latitudes (around 45 degrees) - Lambert Conformal
 - Polar latitudes (more than 75 degrees) - Polar Stereographic
 - Low latitudes (up to 30 degrees and crossing the equator) - Mercator
 - Crossing more than 45 degrees extent in latitude - Rotated Mercator
2. The model hydrostatic engine does not allow a resolution lower than 20km. If you want a higher resolution consider using the subgridding scheme. ICTP plans to introduce in the future a non-hydrostatic compressible core to the RegCM model.
3. Lowering the top pressure of the model can give you problems in regions with complex topography. Touch the default after thinking twice on that.
4. Always specify `clat` and `clon`, the central domain point, and do fine adjustment of the position moving it around a little bit. A little shift in position and some tests can help you obtain a better representation of coastlines and topography at the coarse resolutions.
5. If using `LAMCON` projection, take care to place the two true latitudes at around one fourth and three fourth of the domain latitude space to better correct the projection distortion of the domain.

6. The pole position for the rotated mercator position should be as near as possible to the center domain position.
7. For the `i_band` parameter, selecting this will enable the tropical band experiment, and the horizontal resolution will be calculated from the number of `jx` points. The projection is set to Normal Mercator, the center of the projection is set to `clat = 0.0`, `clon = 180.0`, and the grid point resolution is calculated as:

$$\frac{2 * \pi * 6370.0}{jx} \quad (6.3)$$

Just remember:

- (a) The model for a tropical band simulation is heavy, as the number of points is usually huge to obtain a good horizontal resolution. Check any memory limit is disabled on your platform before attempting a run.
- (b) The model scales well on a cluster with a large number of processors.

6.1.4 terrainparam namelist

This namelist is used by the `terrain` program to know how you want to generate the `DOMAIN` file. You can control its work using a number of parameters to obtain what you consider the best representation of the physical reality. Do not underestimate what you can do at this early stage, having a good representation of the surface can lead to valuable results later when the model calculates climatic parameters.

```
&terrainparam
domname = 'RegCM',      ! Name of the domain/experiment.
                        ! Controls naming of input files
lresamp = .false.,      ! Do a first resampling before interpolation
smthbdy = .false.,      ! Perform extra smoothing in boundaries
ismthlev = 1,           ! Smoothing level (additional 1-2-1 smoother)
roidem = 1.5,           ! Interpolation radius in ds unit for topography
h2ohgt = .true.,        ! Allow water points to have hgt greater than 0
h2opct = 50.0,          ! Surface min H2O percent to be considered water
lakedpth = .false.,     ! If using lakemod (see below), produce from
                        ! terrain program the domain bathymetry
lsmoist = .false.,      ! Use Satellite Soil Moisture Dataset for
                        ! initialization of soil moisture.
fudge_lnd = .false.,    ! Fudging Control flag, for landuse of grid
fudge_lnd_s = .false.,  ! Fudging Control flag, for landuse of subgrid
fudge_tex = .false.,    ! Fudging Control flag, for texture of grid
fudge_tex_s = .false.,  ! Fudging Control flag, for texture of subgrid
fudge_lak = .false.,    ! Fudging Control flag, for lake of grid
fudge_lak_s = .false.,  ! Fudging Control flag, for lake of subgrid
dirter = 'input/',      ! Output directory for terrain files
inpter = 'globdata/',   ! Input directory for SURFACE dataset
tersrc = 'GMTED',       ! Select GMTED or GTOPO DEM data
smsrc = 'ESACCI',       ! Select ESACCI or CPC surface soil moisture
                        ! when lsmoist option is True
moist_filename = 'moist.nc', ! Read initial moisture and snow from this file
/
```

The things you need to know here:

1. The `domname` will control the output file naming convention, all generated files will add this prefix to the old V3 naming convention, giving you the capability to recognize different runs. Try to use always meaningful names.
2. You can control the final land-water mask using the `h2opct` parameter. This parameter can be used to have more land points than calculated by the simple interpolation engine. Try it with different values to find best land shapes. A zero value means use just the interpolation engine, higher values will extend into ocean points the land at land-water interface. The `h2ohgt` parameter allows also water points to have elevation greater than zero to avoid wall effects on the coasts.

3. A number of flags control the capability of the `terrain` program to modify on request the class type variables in the `DOMAIN` file. You can modify on request the `landuse`, the texture and the lake/land interface. Running once the `terrain` program, it will generate for you aside from the `DOMAIN` file a series of ASCII files you can modify with any text editor. Running the `terrain` program the second time and setting a `fudge` flag, will tell the program to overwrite the selected variable with the modified value in the ASCII file. This can be useful for sensitivity experiments in the BATS surface model or to design a scenario experiment.
4. Some of the land surface types in BATS have been little tested and used or are extremely simplified and thus should be used cautiously. Specifically the types are: sea ice, bog/marsh, irrigated crop, glacier. If such types are present in a domain, the user is advised to carefully check the model behavior at such points and eventually substitute these types with others.
5. The `inpter` directory is expected to contain a `SURFACE` directory where the actual netCDF global dataset are stored. The overall path is limited to 256 characters.
6. If the netCDF library is compiled with OpenDAP support, an URL can be used as a path in the `dirter` and `inpter` variables. Note that the 256 character limit for paths holds in the whole program.
7. The `lsmoist` namelist entry triggers the interpolation on the domain area of a global dataset of satellite measured surface soil moisture to estimate the initial model soil moisture. A simple algorithm extend the surface soil moisture to all the soil layers.
8. The `moist_filename` if present triggers reading from the specified file the soil moisture on all model vertical layers and use it to initialize the soil moisture content in an initial run. Note that for this a `DOMAIN` file created for a CLM45 run cannot be used for a BATS run.

6.1.5 globdatparam namelist

This namelist is used by the `sst` and `icbc` ICBC programs. You can tell them how to build initial and boundary conditions.

```
!
! ICBC Global data input control
!
!   GLOBAL DATA AVAILABLE FROM ICTP FROM:
!       http://clima-dods.ictp.it/Data/RegCM_Data
!
&globdatparam
ibdyfrq =      6,          ! boundary condition interval (hours)
ssttyp = 'EIN75',         ! Type of Sea Surface Temperature used
!       One in: GISST, OISST, OI2ST, OI_WK, OI2WK,
!               FV_A2, FV_B2, EH5A2, EH5B1, EHA1B,
!               EIN75, EIN15, CCSST, CA_XX, HA_XX,
!               EC_XX, IP_XX, GF_XX, CN_XX, MP_XX
dattyp = 'EIN75',         ! Type of global analysis datasets used
!       One in: ECMWF, ERA40, EIN75, EIN15, EIN25,
!               ERAHI, NNRP1, NNRP2, NRP2W, GFS11,
!               FVGCM, FNEST, EH5A2, EH5B1, EHA1B,
!               CCSMN, ECEXY, CA_XX, HA_XX, EC_XX,
!               IP_XX, GF_XX, CN_XX, MP_XX
!       with XX for CMIP5 datasets in 26, 45, 85
chemtyp = 'MZCLM',        ! Type of Global Chemistry boundary conditions
!       One in : MZ6HR, 6 hours MOZART output
!               : MZCLM, MOZART climatology 1999-2009
gdate1 = 1990060100,      ! Start date for ICBC data generation
gdate2 = 1990070100,      ! End data for ICBC data generation
calendar = 'gregorian',   ! Calendar type : gregorian, noleap, 360_day
dirglob = 'input',        ! Path for ICBC produced input files
inglob = 'RegCM_Data',    ! Path for ICBC global input datasets.
ensemble_run = .false.,   ! If this is a member of a perturbed ensemble
! run. Activate random noise added to input
! ICBC controlled by the perturbparam stanza
/
```

Things you need to know here:

1. The `gdate` time window to build ICBC must be always greater or equal to the time window you plan to run the model in. Different GCMs and reanalysis products have different length of the year. For example, the reanalysis products employ the real year length (365 days + real leap years, i.e. and average length of 365.2422), the CCSM has a length of 365 days (no leap year), the HadCM has a length of 360 days (30 day months). The RegCM4 length of the year has to be the same as in the forcing fields, and this can be set in the variable `dayspy`. Please remember to always check the consistency of the length of the year.
2. Even if listed, not all the input engines are fully tested. Some of them need data which have been reformatted by ICTP (they are not in the original format with which they are distributed by the institution producing them). Some input data are not freely distributable by ICTP, and you need a special agreement with the owner to use them. Hopefully the situation is changing, and data exchange is becoming more and more the basis for good science in the climatic field.
3. The `chemtyp` parameter is read by the `chem_icbc` program. See below in 6.1.25 about chemistry boundary conditions.
4. For notes on path, you can see the above in `terrainparam` namelist description at 5.

6.1.6 `fnestparam` namelist

This namelist is read if the `FNEST` is selected as `dattyp` in `globdatparam` namelist (see above in 6.1.5), and permits the user to specify the output directory of the coarse resolution run already completed and the name of the coarse domain, i.e. the `domname` namelist parameter used in the `terrainparam` coarse namelist (see above in 6.1.4).

The nested domain must be contained inside the coarse domain, and possibly the nested domain should not overlap the boundary region of the coarse run. Those checks are left to the user.

If nothing is specified or namelist not present default is to search for a directory `RegCM` inside `inpglob` directory for files without a `domname`, i.e. like `ATM.YYYYMMDDHH.nc`.

```
!  
! Nesting control  
!  
&fnestparam  
coarse_outdir = 'globdata/RegCM', ! Coarse domain output dir if FNEST  
coarse_domname = 'EUROPE',        ! Coarse domain domname  
/
```

6.1.7 `restartparam` namelist

This namelist lets you control the time period the model is currently simulating in this particular run. You may want to split longer runs for which you have prepared the ICBC's into shorter runs, to schedule HPC resource usage in a more collaborative way with other researcher sharing it: the `regcm` model allows restart, so be friendly with other research projects which may not have this fortune (unless you are late for publication).

```
&restartparam  
ifrest = .false., ! If a restart  
mdate0 = 1990060100, ! Global start (is gdate1, most probably)  
mdate1 = 1990060100, ! Start date of this run  
mdate2 = 1990060200, ! End date for this run  
/
```

Things you need to know here:

1. After the simulation starts, on restart NEVER change the `mdate0` value. The correct scheme for restart is:
 - Set `ifrest` to `.true.`
 - Set `mdate1` to the value in `mdate2`

- Define the new value for `mdate2`
2. Consider that current RegCM convention is to place midnight of first day of month as the last timestep in previous month, except on first model output file (`ifrest = .false.`). It is for this reason better to use as start and end time a month boundary. We usually consider a month data file the basic unit of output, each time you cross a month a new output file will be created for you.

6.1.8 timeparam namelist

This namelist contains model internal timesteps, used by the model as basic integration timestep and triggers for calling internal parametric schemes.

```
&timeparam
dt      = 150., ! time step in seconds
dtrad   = 0., ! time interval solar radiation calculated (minutes)
dtsrf   = 0., ! time interval at which land model is called (seconds)
dtcum   = 0., ! time interval at which cumuls is called (seconds)
dtabem  = 0., ! time interval absorption-emission calculated (hours)
dtchem  = 0., ! time interval for chemistry reactions (seconds)
/
```

Things you need to know here:

1. If only `dt` is chosen, the other values are computed to align with `dt` and output frequencies to have *5minutes* for cumulus, *10minutes* for surface, *30minutes* for radiation, *18hr* for absorption-emission.
2. All the internal timesteps need to be multiples of the base timestep `dt`. Note that the units are different, so you need to convert the other timesteps in seconds before the check.
3. Surface, radiation and cumulus timesteps must be aligned on a *24h* time window.
4. The dynamical hydrostatical core of RegCM requires a fixed timestep, and you need to manually find the correct value which permits not to break the Courant–Friedrichs–Lewy condition considering *R. Courant and Lewy (1928)*. A good rule of thumb is to have a `dt` not greater than three times the `ds` value in *km* specified in the `geoparam` namelist at 6.1.3. A greater value may lower computing time, but in case of strong advection may lead to non accurate computation or even the violation of CFL condition and the divergence of the solution.
5. If you hit a non stable condition, the restart capability of the model may help find the correct timestep just for a particular period, using a different timestep at different times.
6. The surface model timestep should be of the order of *10minutes* to keep computational cost low: lower values may be used only in the case of very strong surface gradients.
7. A negative value for `dtcum` (the default), sets `dtcum = dt`, calling the cumulus convection scheme every model timestep. The user selected value can be reasonably be any multiple of the `dt` value up to the maximum value of *5minute* to reduce computational load. In case of very fast and intense convection, safest option is to use the dynamical core timestep.
8. The absorption-emission computations are the most expensive in computational time in the radiation scheme, which accounts overall for around 30% of the total execution time. The *18hours* default allows for a compromise for accuracy/cost and should avoid aliasing on a daily timeframe window.

6.1.9 outparam namelist

This namelist controls the model output engine, allowing you to enable/disable any of the output file writeout, or to modify the frequency the fields are written in the files.

```

!
! Model Output control
!
&outparam
outnwf = 0., ! Day interval to open new files (0 = monthly)
ifsave = .true., ! Create SAV files for restart
savfrq = 0., ! Frequency in days to create them (0 = monthly)
ifatm = .true., ! Output ATM ?
atmfrq = 6., ! Frequency in hours to write to ATM
ifrad = .true., ! Output RAD ?
radfrq = 6., ! Frequency in hours to write to RAD
ifsrf = .true., ! Output SRF ?
srffrq = 3., ! Frequency in hours to write to SRF
ifsts = .true., ! Output STS (frequency is daily) ?
ifshf = .true., ! Output SHF (frequency is hourly) ?
ifsub = .true., ! Output SUB ?
subfrq = 6., ! Frequency in hours to write to SUB
iflak = .true., ! Output LAK ?
lakfrq = 6., ! Frequency in hours to write to LAK
ifchem = .true., ! Output CHE ?
ifopt = .false., ! Output OPT ?
chemfrq = 6., ! Frequency in hours to write to CHE
enable_atm_vars = 73*.true., ! Mask to eventually disable variables ATM
enable_srf_vars = 46*.true., ! Mask to eventually disable variables SRF
enable_rad_vars = 28*.true., ! Mask to eventually disable variables RAD
enable_sub_vars = 18*.true., ! Mask to eventually disable variables SUB
enable_sts_vars = 18*.true., ! Mask to eventually disable variables STS
enable_shf_vars = 8*.true., ! Mask to eventually disable variables SHF
enable_lak_vars = 18*.true., ! Mask to eventually disable variables LAK
enable_opt_vars = 20*.true., ! Mask to eventually disable variables OPT
enable_che_vars = 26*.true., ! Mask to eventually disable variables CHE
dirout = './output', ! Path where all output will be placed
lsync = .false., ! If sync of output files at every timestep is
! requested. Note, it has a performance impact.
! Enabled by default if debug_level > 2
uvrotate = .false., ! Rotate projected wind to S-N, W-E direction
icosp = 0, ! Enable COSP needed fields in output in the ATM
idiag = 0, ! Enable tendency diagnostic output in the ATM
! file. NOTE: output file gets HUGE.
do_parallel_netcdf_in = .false., ! This enables parallel input
! Each processors reads its slice in the
! input file. Enable ONLY in case of
! HUGE input bandwidth,
do_parallel_netcdf_out = .false., ! This enables parallel output if the
! hdf5/netcdf libraries support it and
! the model is compiled with :
! --enable-nc4-parallel
deflate_level = 1, ! Deflate level for NetCDF4 compress format if model is
! compiled with both --enable-nc4 and --enable-nc4-gzip
/

```

Things you need to know here:

1. The surface fields are the mean values in the interval specified by the frequency values. The dynamical fields are instead the point value at the output time. Refer to the Reference Manual *Giorgi (2011)* for a detailed description of the model output fields.
2. If the chemistry or lake model are not enabled, the values specified in the control flags are not considered. If `nsg` is not greater than one in `dimparam` at 6.1.2, the `ifsub` flag is not considered.
3. For the output directory, the path variable has a limit of 256 characters. This path must be a local path on disk where the user running the model has write permissions granted.
4. The `enablevar` logical arrays can be used to avoid saving one of the time dependent variables in the output file, in the order they are saved in the output file itself. Note that geolocation and pressure variables cannot be disabled.

6.1.10 physicsparam namelist

This namelist controls the model physics. You have a number of option here, and the best way to select the right set is to carefully read the the Reference Manual *Giorgi* (2011). We are for the purposes of this User Guide not going in detail in here, except in saying that probably you will need to run some experiments especially with different cumulus convection schemes before finding out the best model setting. Although the mixed convection scheme (Grell over land and Emanuel over ocean) seems to provide an overall better performance, our experience is that there is no scheme that works best everywhere, therefore we advice to always do some sensitivity experiments to select the best scheme for your application.

```
&physicsparam
iboudy =      5, ! Lateral Boundary conditions scheme
               ! 0 => Fixed
               ! 1 => Relaxation, linear technique.
               ! 2 => Time-dependent
               ! 3 => Time and inflow/outflow dependent.
               ! 4 => Sponge (Perkey & Kreitzberg, MWR 1976)
               ! 5 => Relaxation, exponential technique.
isladvec =    0, ! Semilagrangian advection scheme for tracers and
               ! humidity
               ! 0 => Disabled
               ! 1 => Enable Semi Lagrangian Scheme
iqmsl =       1, ! Quasi-monotonic Semi Lagrangian
               ! 0 => Standard Semi-Lagrangian
               ! 1 => Bermejo and Staniforth 1992 QMSL scheme
ibltyp =      1, ! Boundary layer scheme
               ! 0 => Frictionless
               ! 1 => Holtslag PBL (Holtslag, 1990)
               ! 2 => UW PBL (Bretherton and McCaa, 2004)
               ! 3 => GFS 2011
               ! 4 => MYJ
idiffu =      1, ! Diffusion scheme
               ! 0 => No diffusion
               ! 1 => MM5 4th order interior/ 2nd order boundary
               ! 2 => LeVeque 4th order 9 point laplacian
               ! 3 => Xue 6th order with flux limiter
icup_lnd =    4, ! Cumulus convection scheme Over Land
icup_ocn =    4, ! Cumulus convection scheme Over Icean
               ! 1 => Kuo
               ! 2 => Grell
               ! 3 => Betts-Miller (1986) DOES NOT WORK !!!
               ! 4 => Emanuel (1991)
               ! 5 => Tiedtke (1996)
               ! 6 => Kain-Fritsch (1990), Kain (2004)
               ! -1 => MM5 Shallow cumulus scheme:
               !       No precipitation but only mixing.
ipptls =      1, ! Moisture scheme
               ! 1 => Explicit moisture (SUBEX; Pal et al 2000)
               ! 2 => Explicit moisture Nogherotto/Tompkins
               ! 3 => Explicit moisture WSM5
iocncpl =     0, ! Ocean SST from coupled Ocean Model through RegESM
               ! 1 => Coupling activated
iwavcpl =     0, ! Ocean roughness from coupled Wave Model through RegESM
               ! 1 => Coupling activated
icopcpl =     0, ! Export data for COP component (RegESM)
               ! 1 => Coupling activated
iocnflx =     2, ! Ocean Flux scheme
               ! 1 => Use BATS1e Monin-Obukhov
               ! 2 => Zeng et al (1998)
               ! 3 => Coare bulk flux algorithm
iocnrough =   1, ! Zeng Ocean model roughness formula to use.
               ! 1 => (0.0065*ustar*ustar)/egrav
               ! 2 => (0.013*ustar*ustar)/egrav + 0.11*visa/ustar
               ! 3 => (0.017*ustar*ustar)/egrav
               ! 4 => Huang 2012 free convection and swell effects
```

```

! 5 => four regime formulation
iocnzoq = 1, ! Zeng Ocean model factors for t,q roughness
! 1 => 2.67*(re**d_rfou) - 2.57
! 2 => min(4.0e-4, 2.0e-4*re**(-3.3))
! 3 => COARE formulation as in bulk flux above
ipgfg = 0, ! Pressure gradient force scheme
! 0 => Use full fields
! 1 => Hydrostatic deduction with pert. temperature
iemiss = 0, ! Use computed long wave emissivity
lakemod = 0, ! Use lake model
ichem = 0, ! Use active aerosol chemical model
scenario = 'RCP4.5', ! AR5 RCP scenario in RCP2.6, RCP4.5, RCP6.0, RCP8.5
! AR4 old CMIP3 scenario to use in A1B, A2, B1, B2
! CONST scenario at year ghg_year_const
ghg_year_const = 1950, ! Year to use for a constant GHG concentration values
idcsst = 0, ! Use diurnal cycle sst scheme
ipcpcool = 0, ! Precipitation cooling effect over ocean surface
iwhitecap = 0, ! Wind whitecapping effect on ocean albedo
iseaice = 0, ! Model seaice effects
idesseas = 0, ! Model desert seasonal albedo variability
iconvlpw = 1, ! Use convective algo for lwp in the large-scale
! This is reset to zero if using ipptls = 2
icldfrac = 0, ! Cloud fraction algorithm
! 0 : Original SUBEX
! 1 : Xu-Randall empirical
! 2 : Thompson scheme
! 3 : Gultepe-Isaac scheme
icldmstrat = 1, ! Simulate stratocumulus clouds
icumcloud = 1, ! Kuo and Betts Miller formula to use for cumulus
! clouds (cf and lwc) cloud fractions
! It is only used only if mass fluxes are not
! available (Kuo and BM).
! 0,1 => cf = 1-(1-clfrcv)**(1/kdepth)
! 2 => cf = cloud profile
! Liquid water content:
! 0 => constant in cloud
! 1,2 => function of temperature
irrtm = 0, ! Use RRTM radiation scheme instead of CCSM
iclimao3 = 0, ! Use O3 climatic dataset from SPARC CMIP5
iclimaaer = 0, ! Use AEROSOL climatic dataset from AERGLOB for non
! interactive aerosol load affecting radiative scheme.
! Requires running chem_icbc
isolconst = 0, ! Use a constant 1367 W/m^2 instead of the prescribed
! TSI recommended CMIP5 solar forcing data.
islab_ocean = 0, ! Activate the SLAB ocean model
itweak = 0, ! Enable tweak scenario
ifixsolar = 0, ! Fix the solar constant to fixedsolarval
! (no diurnal or seasonal cycle)
fixedsolarval = 343., ! The constant solar value for ifixsolar = 1
/

```

6.1.11 dynparam namelist

This namelist controls some fine tuning parameters for the dynamical cores.

```

!
! Dynamical core parameters : Use defaults, be on your own otherwise!
!
&dynparam
gnu1 = 0.0625, ! nu factor for Asselin filter in leapfrog step.
gnu2 = 0.0625, ! nu factor for Asselin filter in leapfrog step (tracers).
! MM5 manual , equation 2.4.6
! Default 0.0625 for hydro, 0.1 for nonhydro
ckh = 1.0, ! Background diffusion multiplication factor
adyndif = 1.0, ! Dynamical diffusion multiplication factor
diffu_hgtf = 1, ! Add topographic effect to diffusion

```

```

! It is set to zero default in the non-hydrostatic
upstream_mode = .true., ! Add off centering to advection
uoffc = 0.250, ! Maximum off-centering factor to use
stability_enhance = .true., ! Do not allow horizontal advection to increase
! tendencies if strong gradients are present
t_extrema = 5.0, ! Maximum gradient of T in K for advection to stop
q_rel_extrema = 0.2, ! Maximum gradient fraction for QV for advection to stop
/

```

Best leave this parameters as they are if you do not have problematic instability issues.

6.1.12 referenceatm and nonhydroparam namelist

These namelist controls non-hydrostatic core for the upper radiative boundary conditions and the base state vertical profile curve shape.

```

&referenceatm
base_state_pressure = 101325.0, ! Base state reference pressure
logp_lrate = 47.70, ! Logp lapse rate d(T)/d(ln P) [K/ln(Pa)]
/
&nonhydroparam
ifupr = 1, ! Upper radiative boundary condition (Klemp and Durran,
! Bougeault, 1983)
nhbet = 0.4, ! Ikawa beta parameter (0.=centered, 1.=backward)
! determines the time-weighting, where zero gives a
! time-centered average and positive values give a bias
! towards the future time step that can be used for
! acoustic damping. In practice, values of
! nhbet = 0.2 - 0.4 are used (MM5 manual, Sec. 2.5.1)
nhxkd = 0.1, ! Time weighting for weighting old/new pp
ifrayd = 1, ! Upper levels Rayleigh damper to BCs
rayndamp = 5, ! Number of top levels to apply
rayalpha0 = 0.001, ! Rate alpha0
rayhd = 10000.0, ! Damping scale depth
/

```

1. The upper radiative boundary layer option has a high computational cost.

6.1.13 hydroparam namelist

These namelist controls hydrostatic core split explicit scheme for fast wave removal. User is not advised to change the default values.

```

!
! Hydrostatic core option
!
&hydroparam
nsplit = 2, ! Number of split explicit timesteps
lstand = .true., ! Use standard atmosphere for vertical modes linearization
/

```

6.1.14 boundaryparam namelist

Being a limited area model, in order to be run RegCM4 requires the provision of meteorological initial and time dependent lateral boundary conditions, typically for wind components, temperature, water vapor and surface pressure. These are obtained by interpolation from output from reanalysis of observations or global climate model simulations, which thus drive the regional climate model.

The lateral boundary conditions (LBC) are provided through the so called relaxation/diffusion technique which consists of:

1. selecting a lateral buffer zone of n grid point width (nspgx)
2. interpolating the driving large scale fields onto the model grid

3. applying the relaxation + diffusion term

$$\frac{\partial \alpha}{\partial t} = F(n)F_1 * (\alpha_{LBC} - \alpha_{mod}) - F(n)F_2 * \Delta_2(\alpha_{LBC} - \alpha_{mod}) \quad (6.4)$$

where α is a prognostic variable (wind components, temperature, water vapor, surface pressure). The first term on the rhs is a Newtonian relaxation term which brings the model solution (*mod*) towards the LBC field (*LBC*) and the second term diffuses the differences between model solution and LBC. $F(n)$ is an exponential function given by:

$$F(n) = \exp\left(\frac{-(n-1)}{anudge(k)}\right) \quad (6.5)$$

Where n is the grid point distance from the boundary (varying from 1 to *nspgx*): $n-1$ is the outermost grid point, $n=2$ the adjacent one etc. The *anudge* array determines the strength of the LBC forcing and depends on the model level k . In practice $F(n)$ is equal to 1 at the outermost grid point row and decreases exponentially to 0 at the internal edge of the buffer zone (*nspgd*) at a rate determined by *anudge*. Larger buffer zones and larger values of *anudge* will yield a greater forcing by the LBC.

Typically for domain sizes of 100 grid points we use a buffer zone width of 10 – 12 grid points, for large domains this buffer zone can increase to values of 15 or even 20.

In the model *anudge* has three increasing values from the lower, to the mid and higher troposphere. For example for *nspgx* = 10 we use *anudge* equals to 1, 2, 3 for the lower, mid and upper troposphere, respectively.

This allows a stronger forcing in the upper troposphere to insure a greater consistency of large scale circulations with the forcing LBC while allowing more freedom to the model in the lower troposphere where local high resolution forcings (e.g. complex topography) are more important.

For *nspgx* of 15 – 20, for example, *anudge* values could be increased to 2, 3, 4 or even 2, 4, 6. As a rule of thumb, the choice of the number of *nspgx* points and the maximum *anudge* value should follow the conditions:

$$2 \text{ } nspgx < 0.25 \min(IY, JX) \quad (6.6)$$

$$anudge \leq \frac{(nspgx - 1)}{3} \quad (6.7)$$

Most notably to have roughly the same buffer zone in doubling the resolution, you should double the number of *nspgx* points and increase up to double the value of *anudge*.

```
&boundaryparam
nspgx = 12, ! nspgx-1 represent the number of cross point slices on
           ! the boundary sponge or relaxation boundary conditions.
nspgd = 12, ! nspgd-1 represent the number of dot point slices on
           ! the boundary sponge or relaxation boundary conditions.
high_nudge = 3.0, ! Nudge value high range
medium_nudge = 2.0, ! Nudge value medium range
low_nudge = 1.0 ! Nudge value low range
bdy_nm = -1.0, ! Newtonian term, Eq. 7 Giorgi et al, 1993
           ! Default is to use the formulation 1/dt
bdy_dm = -1.0, ! Reverse of diffusion term, Eq. 8 Giorgi et al, 1993
           ! Default is to use the formulation 1/(50*dt)
/
```

6.1.15 cldparam namelist

This namelist controls the cloud fraction algorithms.

```
&cldparam
ncld = 1, ! # of bottom model levels with no clouds (rad only)
rhmax = 1.01, ! RH at which FCC = 1.0
rhmin = 0.01, ! RH min value
rh0land = 0.80, ! Relative humidity threshold for land (subex)
rh0oce = 0.90, ! Relative humidity threshold for ocean (subex)
tc0 = 238.0, ! Below this temp, rh0 begins to approach unity (subex)
```

```

cllwcv   = 0.3e-3, ! Cloud liquid water content for convective precip.
clfrvcmax = 0.75, ! Max cloud fractional cover for convective precip.
cftotmax = 0.75, ! Max total cover cloud fraction for radiation
k2_const  = 10.0, ! K2 Factor cumulus mass flux - cloud fraction
kfac_shal = 0.07, ! Factor cumulus mass flux - cloud fraction - shallow
kfac_deep = 0.14, ! Factor cumulus mass flux - cloud fraction - deep
lsrfhack  = .false. ! Surface radiation hack
larcticcorr = .true. ! Vavrus and Waliser Arctic cloud correction
rcrit     = 13.5, ! Mean critical radius !
coef_ccn  = 2.5e+20, ! Coefficient determined by assuming a lognormal PMD
abulk     = 0.9, ! Bulk activation ratio
/

```

6.1.16 subexparam namelist

This namelist controls the SUBEX moisture scheme. Please consider carefully reporting in your work the tuning you perform on this parameters. The parameters below are the ones currently used at ICTP.

```

&subexparam
qcklland = 0.0005, ! Autoconversion Rate for Land
qckloce  = 0.0005, ! Autoconversion Rate for Ocean
gulland  = 0.65, ! Fract of Gultepe eqn (qcth) when prcp occurs (land)
guloce   = 0.30, ! Fract of Gultepe eqn (qcth) for ocean
cevaplnd = 1.0e-5, ! Raindrop evap rate coef land [[(kg m-2 s-1)-1/2]/s]
cevapoce = 1.0e-5, ! Raindrop evap rate coef ocean [[(kg m-2 s-1)-1/2]/s]
cacclnd  = 6.0, ! Raindrop accretion rate land [m3/kg/s]
caccroce = 4.0, ! Raindrop accretion rate ocean [m3/kg/s]
conf     = 1.00, ! Condensation efficiency
/

```

We found that RegCM4 is especially sensitive to:

1. cevap : increasing cevap will generally decrease precipitation
2. gulland, guloce : increase of guland/guloce will generally lead to reduce precipitation

6.1.17 microparam namelist

This namelist controls the new microphysics scheme.

```

&microparam
stats = .false., ! Produce debug variables in output files
budget_compute = .false., ! Verify enthalpy and moisture conservation
nssopt = 1, ! Supersaturation Computation
! 0 => No scheme
! 1 => Tompkins
! 2 => Lohmann and Karcher
! 3 => Gierens
iautoconv = 4, ! Choose the autoconversion paramaterization
! => 1 Klein & Pincus (2000)
! => 2 Khairoutdinov and Kogan (2000)
! => 3 Kessler (1969)
! => 4 Sundqvist
vfqr = 4.0, ! Rain fall speed (default is 4 m/s)
vfqi = 0.15, ! Ice fall speed (default is 0.15 m/s)
vfqs = 1.0, ! Snow fall speed (default is 1 m/s)
auto_rate_khair = 0.355, ! Autoconversion coefficient for iautoconv=2
auto_rate_kessl = 1.e-3, ! Autoconversion coefficient for iautoconv=3
auto_rate_klepi = 0.5e-3, ! Autoconversion coefficient for iautoconv=1
rkconv = 1.666e-4, ! Autoconversion coefficient for iautoconv=4
skconv = 1.0-3, ! Autoconversion coefficient for snow
rcovpmin = 0.1, ! Minimum precipitation coverage
rpecons = 5.547e-5, ! Evaporation constant Kessler
/

```

6.1.18 grellparam, emanparam, tiedtkeparam and kfparam namelists

You are allowed here to tune the convection scheme selected above in 6.1.10 with the icup_lnd or icup_ocn number if selected number is 2,4,5.

```
&grellparam
igcc = 2,          ! Cumulus closure scheme
                  ! 1 => Arakawa & Schubert (1974)
                  ! 2 => Fritsch & Chappell (1980)
edtmn            = 0.20, ! Minimum Precipitation Efficiency land
edtmn_ocn        = 0.20, ! Minimum Precipitation Efficiency ocean
edtmax           = 0.80, ! Maximum Precipitation Efficiency land
edtmax_ocn       = 0.80, ! Maximum Precipitation Efficiency ocean
edtmino          = 0.20, ! Minimum Tendency Efficiency (o var) land
edtmino_ocn      = 0.20, ! Minimum Tendency Efficiency (o var) ocean
edtmaxo          = 0.80, ! Maximum Tendency Efficiency (o var) land
edtmaxo_ocn      = 0.80, ! Maximum Tendency Efficiency (o var) ocean
edtminx          = 0.20, ! Minimum Tendency Efficiency (x var) land
edtminx_ocn      = 0.20, ! Minimum Tendency Efficiency (x var) ocean
edtmaxx          = 0.80, ! Maximum Tendency Efficiency (x var) land
edtmaxx_ocn      = 0.80, ! Maximum Tendency Efficiency (x var) ocean
shrmin           = 0.30, ! Minimum Shear effect on precip eff. land
shrmin_ocn       = 0.30, ! Minimum Shear effect on precip eff. ocean
shrmax           = 0.90, ! Maximum Shear effect on precip eff. land
shrmax_ocn       = 0.90, ! Maximum Shear effect on precip eff. ocean
pbcmax           = 100.0, ! Max depth (mb) of stable layer b/twn LCL & LFC
mincld           = 50.0,  ! Min cloud depth (mb).
htmin            = -250.0, ! Min convective heating
htmax            = 500.0,  ! Max convective heating
skbmax           = 0.4,   ! Max cloud base height in sigma
dtauc            = 60.0,  ! Fritsch & Chappell (1980) ABE Removal Timescale (min)
/

&emanparam
minsig = 0.95,      ! Lowest sigma level from which convection can originate
elcrit_ocn = 0.0011, ! Autoconversion threshold water content (g/g) (ocean)
elcrit_lnd = 0.0011, ! Autoconversion threshold water content (g/g) (land)
tlcrit = -55.0,     ! Below tlcrit auto-conversion threshold is zero
entp = 0.06,        ! Coefficient of mixing in the entrainment formulation
sigd = 0.05,        ! Fractional area covered by unsaturated dndraft
sigs = 0.15,        ! Fraction of precipitation falling outside of cloud
omtrain = 50.0,     ! Fall speed of rain (Pa/s)
omtsnow = 5.5,      ! Fall speed of snow (Pa/s)
coeffr = 1.0,       ! Coefficient governing the rate of rain evaporation
coeffs = 0.8,       ! Coefficient governing the rate of snow evaporation
cu = 0.7,           ! Coefficient governing convective momentum transport
betae = 10.0,       ! Controls downdraft velocity scale
dtmax = 0.65,       ! Max negative parcel temperature perturbation below LFC
alphae = 0.02,      ! Controls the approach rate to quasi-equilibrium
damp = 0.01,        ! Controls the approach rate to quasi-equilibrium
epmax_ocn = 0.999,  ! Maximum precipitation efficiency (ocean)
epmax_lnd = 0.999,  ! Maximum precipitation efficiency (land)
/

&tiedtkeparam
iconv = 4,          ! Actual used scheme.
entrmax = 1.75e-3,  ! Max entrainment iconv=[1,2,3]
entrdd = 3.0e-4,    ! Entrainment rate for cumulus downdrafts
entrpen_lnd = 1.75e-3, ! Entrainment rate for penetrative convection land
entrpen_ocn = 1.75e-3, ! Entrainment rate for penetrative convection ocean
entrscv = 3.0e-4,   ! Entrainment rate for shallow convection iconv=[1,2,3]
entrmid = 1.0e-4,   ! Entrainment rate for midlevel convection iconv=[1,2,3]
cprcon = 1.0e-4,    ! Coefficient for determining conversion iconv=[1,2,3]
detrpen_lnd = 0.75e-4, ! Detrainment rate for penetrative conv land iconv=4
detrpen_ocn = 0.75e-4, ! Detrainment rate for penetrative conv ocean iconv=4
entshalp = 2.0,     ! shallow entrainment factor for entrorg iconv=4
rcuc_lnd = 0.05,    ! Convective cloud cover for rain evaporation iconv=4
```



```

rcuc_ocn = 0.05,      ! Convective cloud cover for rain evaporation iconv=4
rcpec_lnd = 5.55e-5, ! Coefficient for rain evaporation below cloud iconv=4
rcpec_ocn = 5.55e-5, ! Coefficient for rain evaporation below cloud iconv=4
rhebc_lnd = 0.8,      ! Critical rh below cloud for evaporation iconv=4
rhebc_ocn = 0.8,      ! Critical rh below cloud for evaporation iconv=4
rprc_lnd = 1.4e-3,    ! conversion coefficient from cloud water iconv=4
rprc_ocn = 1.4e-3,    ! conversion coefficient from cloud water iconv=4
cmtcape = 3600.0,     ! CAPE adjustment timescale iconv=[1,2,3]
lmfpen  = .true.,     ! penetrative conv is switched on
lmfmid  = .true.,     ! midlevel conv is switched on
lmfdd   = .true.,     ! cumulus downdraft is switched on
lepcl   = .true.,     ! prognostic cloud scheme is on
lmfdudv = .true.,     ! cumulus friction is switched on
lmfscv  = .true.,     ! shallow convection is switched on
lmfuvdis = .true.,    ! use kinetic energy dissipation
lmftrac  = .true.,    ! chemical tracer transport is on
lmfsmooth = .false.,  ! smoot of mass fluxes for tracers
lmfwstar = .false.,   ! Grant w* closure for shallow conv
/

&kfparam
kf_entrates = 0.03,      ! Entrainment rate
kf_convrates = 0.03,     ! Condensate to rain conversion rate
kf_min_pecf = 0.2,       ! Minimum precipitation efficiency
kf_max_pecf = 0.9,       ! Maximum precipitation efficiency
kf_dpp      = 150.0,     ! Start elevation for downdraft above cloud base (mb)
kf_tkemax   = 5.0,       ! Maximum turbulent kinetic energy in sub cloud layer
kf_min_dtcape = 1800.0,  ! Consumption time of CAPE low limit
kf_max_dtcape = 3600.0,  ! Consumption time of CAPE high limit
/

```

Things you need to know here:

1. In case of mixed cumulus schemes (land/ocean), both the selected schemes configuration namelist are read in. Note in this case for the schemes only the relevant (Ocean or Land) control values are used.
2. Minimum and maximum values of the fraction of reevaporated water in the downdraft for the Grell scheme is essentially a measure of the precipitation efficiency: increasing their value generally decrease convective precipitation.
3. Again, read carefully the Reference Manual before attempting any tuning, and report in any work modification of this parameters.

6.1.19 holtslagparam namelist

You are allowed here to tune the Holtslag PBL scheme selected above in 6.1.10 with the `ibltyp` number if selected number is 1.

```

&holtslagparam
ricr_ocn = 0.25, ! Critical Richardson Number over Ocean
ricr_lnd = 0.25, ! Critical Richardson Number over Land
zhnew_fac = 0.25, ! Multiplicative factor for zzhnew in holtpbl
ifaholtth10 = 1, ! First approximation for obhukov length, th10 formula:
!      0 => 0.5 * (t+tg) * (1+0.61*q)
!      1 => (0.25*t + 0.75*tg) * (1+0.61*q)
!      2 => theta + hf/(k*us)*log(z/10)
! t = air temp., tg = ground temp., q = ww mix. ratio
! hf = total heat flux, z = elevation
! theta = virt. pot. t
ifaholt = 1, ! th10 final adjustment:
!      0 => no adjustment
!      1 => max(th10,tg)
!      2 => min(th10,tg)
holtth10iter = 1, ! Iterations in th10 computation
/

```

6.1.20 uwparam namelist

You are allowed here to tune the UW PBL scheme selected above in 6.1.10 with the `ibltyp` number if selected number is 2.

```
&uwparam
iuwvadv = 0, ! 0=standard T/QV/QC advection, 1=GB01-style advection
            ! 1 is ideal for MSc simulation, but may have stability issues
atwo = 15.0, ! Efficiency of enhancement of entrainment by cloud evap.
            ! see Grenier and Bretherton (2001) Mon. Wea. Rev.
            ! and Bretherton and Park (2009) J. Clim.
rstbl = 1.5, ! Scaling parameter for stable boundary layer eddy length
            ! scale. Higher values mean stronger mixing in stable
            ! conditions
czero = 5.869, ! Czero constant in UW PBL (eqn 44a and pgs 856-857)
nuk = 5.0, ! Multiplicative factor for diffusion coefficients
/
```

6.1.21 slabocparam namelist

Here you define parameter and stage for the Ocean q-flux adjusted mixed layer model.

```
&slabocparam
do_qflux_adj = .false., ! Activate SLAB Ocean model surface fluxes adjust
                    ! from an already created climatology
do_restore_sst = .true., ! Create during the run the SLAB Ocean model surface
                    ! fluxes climatology to be used in a subsequent run
sst_restore_timescale = 5.0, ! Time interval in days in building the
                    ! q-flux adjustment
mixed_layer_depth = 50.0, ! Depth in meters of the Ocean mixed layer.
/
```

6.1.22 perturbparam namelist

This namelist lets you control to which input field and of what fractional level a perturbation is added at ICBC stage on the input fields. It is read by the ICBC program if the `ensemble_run` parameter in the `globdatparam` namelist is set to true.

```
!
! Perturbation control for ensembles
!
&perturbparam
lperturb_topo = .false., ! Add perturbation to surface elevation
perturb_frac_topo = 0.001, ! Fractional value of the perturbation on topo
lperturb_ts = .false., ! Add perturbation to surface temeprature
perturb_frac_ts = 0.001, ! Fractional value of the perturbation on ts
lperturb_ps = .false., ! Add perturbation to surface pressure
perturb_frac_ps = 0.001, ! Fractional value of the perturbation on ps
lperturb_t = .false., ! Add perturbation to temperature
perturb_frac_t = 0.001, ! Fractional value of the perturbation on t
lperturb_q = .false., ! Add perturbation to humidity mixing ratio
perturb_frac_q = 0.001, ! Fractional value of the perturbation on q
lperturb_u = .false., ! Add perturbation to zonal velocity
perturb_frac_u = 0.001, ! Fractional value of the perturbation on u
lperturb_v = .false., ! Add perturbation to meridional velocity
perturb_frac_v = 0.001, ! Fractional value of the perturbation on v
/
```

Things you need to know here:

1. The `perturb_frac` should not exceed a percent of the field value. The algorithm detail of the applied noise can be found in *O'Brien et al. (2011)*.

6.1.23 tweakparam namelist

This namelist controls the tweaking of the model to obtain custom scenarios, Is enabled if itweak is 1 in 6.1.10.

```
&tweakparam
itweak_sst = 0,          ! Enable adding sst_tweak to input TS
itweak_temperature = 0,  ! Enable adding temperature_tweak to input T
itweak_solar_irradiance = 0, ! Add solar_tweak to solar constant
itweak_greenhouse_gases = 0, ! Multiply gas_tweak_factors to GG concentrations
sst_tweak = 0.0,         ! In K
temperature_tweak = 0.0, ! In K
solar_tweak = 0.0,       ! In W m-2 (1367.0 is default solar)
gas_tweak_factors = 1.0, 1.0, 1.0, 1.0, 1.0,
!                        CO2  CH4   N2O   CFC11 CFC12
```

6.1.24 rrtmparam namelist

You are allowed here to tune the RRTM radiative scheme selected above in 6.1.10 with the irrtrm number if selected number is 1.

```
&rrtmparam
inflagsw = 2, ! 0 = use the optical properties calculated in prep_dat_rrtm
!             (same as standard radiation)
!             ! 2 = use RRTM option to calculate cloud optical properties
!             !             from water path and cloud drop radius
iceflagsw = 3, ! Flag for ice particle specification
!             ! 0 => ice effective radius, r_ec, (Ebert and Curry, 1992),
!             !             r_ec must be >= 10.0 microns
!             ! 1 => ice effective radius, r_ec, (Ebert and Curry, 1992),
!             !             r_ec range is limited to 13.0 to 130.0 microns
!             ! 2 => ice effective radius, r_k, (Key, Streamer Ref. Manual,
!             !             1996), r_k range is limited to 5.0 to 131.0 microns
!             ! 3 => generalized effective size, dge, (Fu, 1996),
!             !             dge range is limited to 5.0 to 140.0 microns
!             !             [dge = 1.0315 * r_ec]
liqflagsw = 1, ! Flag for liquid droplet specification
!             ! 0 => Compute the optical depths due to water clouds in ATM
!             !             (currently not supported)
!             ! 1 => The water droplet effective radius (microns) is input
!             !             and the optical depths due to water clouds are computed
!             !             as in Hu and Stamnes, J., Clim., 6, 728-742, (1993).
inflaglw = 2, ! Flag for cloud optical properties as above but for LW
iceflaglw = 3, ! Flag for ice particle specification as above but for LW
liqflaglw = 1, ! Flag for liquid droplet specification as above but for LW
icld = 1,      ! Cloud Overlap hypothesis
irng = 1,      ! mersenne twister random generator for McICA COH
imcica = 1,    ! Cloud optical depth (extinction) is in cloud quantity
nradfo = 4,    ! ?????????
/
```

6.1.25 chemparam namelist

This namelist controls the chemistry and aerosol options in the RegCM model.

```
&chemparam
chemsimtype = 'CBMZ', ! Which chemical tracers to be activated.
! One in :
!   DUST   : Activate 4 dust bins scheme
!   SSLT   : Activate 2 bins Sea salt scheme
!   DUSS   : Activate DUST + SSLT
!   DU12   : Activate 12 dust bins scheme
!   CARB   : Activate 4 species black/anthropic
!           carbon simulations
!   SULF   : Activate SO2 and SO4 tracers
!   SUCA   : Activate both SUKF and CARB
!   AERO   : Activate all DUST, SSLT, CARB and SULF
```

```

!   CBMZ   : Activate gas phase and sulfate
!   DCCB   : Activate CBMZ +DUST +CARB
!   POLLEN : Activate POLLEN transport scheme
ichsolver = 1, ! Activate the chemical solver
ichsursrc = 1, ! Enable the emissions
ichdrdepo = 1, ! 1 = enable tracer surface dry deposition. For dust,
!             ! it is calculated by a size settling and dry
!             ! deposition scheme. For other aerosol, a dry
!             ! deposition velocity is simply prescribed further.
ichebby = 1, ! Enable boundary conditions read
ichcumtra = 1, ! 1 = enable tracer convective transport and mixing.
ichremlsc = 1, ! 1 = wet removal of chemical species (washout and rainout
!             ! by total rain) is enabled
ichremcvc = 1, ! 1 = wet removal of chemical species (washout and rainout
!             ! by convective rain) is enabled
ichdustemd = 1, ! Choice for parametrisation of dust emission size distribution
!             ! 1 = use the standard scheme (Alfaro et al., Zakey et al.)
!             ! 2 = use the the revised soil granulometry + Kok et al., 2011
!             ! 3 = use the the CLM4.5 dust emission module.
ichdiag = 0, ! 1 = enable writing of additional diagnostics in the output
ismoke = 0, ! Consider emissions from fires (smoke tracer)
idirect = 0, ! Choice to enable or not aerosol feedbacks on radiation and
!             ! dynamics (aerosol direct and semi direct effects):
!             ! 0 = no coupling to dynamic and thermodynamic.
!             ! 1 = no coupling to dynamic and thermodynamic. However
!             !             the clear sky surface and top of atmosphere
!             !             aerosol radiative forcings are diagnosed.
!             ! 2 = allows aerosol feedbacks on radiative,
!             !             thermodynamic and dynamic fields.
iindirect = 0, ! Enable sulfate indirect effect in radiation scheme
ichjphold = 1, ! Impact of cloud aod on photolysis coef
ichlinox = 1, ! LINOX emission considered
ichbion = 0, ! NOx and NH3 emissions from soil manure (Potter et. al 2010)
rdstemfac = 1.0, ! Aerosol correction factor (Laurent et al, 2008)
rocemfac = 1.33, ! OC correction factor to increase emissions for SOA
/

```

The chemsimtype parameter select one in a number of fixed sets which define the nature and number of chemical species and/or transported aerosols, together with which relevant scheme is to be used in the simulation. The implemented possible simulation types for the aerosol/chemistry options are:

1. DUST : Activate 4 dust bins scheme, with on line emission, transport and removal.
2. SSLT : Activate 2 sea salt bins scheme, with on line emission, transport and removal.
3. DUSS : Activate Dust and seasalt scheme, 6 tracers.
4. DU12 : Activate 12 dust bins scheme
5. CARB : Activate 4 species organic and black carbon in both hydrophobic and hydrophilic aerosol scheme, with on line emission, transport and removal.
6. SULF : Activate SO2 and SO4 tracers with simple sulfate oxidation from oxidant climatology, with on line emission, transport and removal.
7. SUCA : Activate both SULF and CARB.
8. AERO : Activate all DUST, SSLT, CARB and SULF
9. CBMZ : Activate CBMZ gas phase only option : 37 tracer are considered here.
10. DCCB : Activate CBMZ +DUST +CARB + sulfate-nitrate-ammonium aerosol calculated with the ISORROPIA gas-aerosol thermodynamical scheme: 50 tracers are considered here.

The more tracers are used, the heavier computationally are the simulations and the outputs. The chemistry outputs consist of one netCDF file per tracer, named explicitly and containing concentration fields + different diagnostics, and one netCDF file giving the optical properties of the total aerosol mixture i.e. aerosol optical depth and radiative forcing. For a big domain, this can require a huge amount of disk space to store the model results.

We will now detail the steps required to run a chemistry/aerosol simulation with the RegCM model.

Pre Processing

We need to perform a couple of operations in the pre-processing stage to prepare input datasets for an aerosol/chemistry simulation.

1. After having prepared the static and boundary condition data with the `icbc` program for the atmosphere, we need also to define chemical emissions and chemical boundary conditions.

The data needed for this second task come from different sources, both measurements data or GCM model with a chemistry parametrization active.

1. Emission dataset. The pre processor can manage CMIP5 RCP and IASA anthropogenic emissions for present day and future emission. For this, the global RCP emission dataset have first been processed by ICTP to match the species considered in the chemical solver CBMZ, and to aggregate different sector of emissions that are present in the RCP fields (e.g. CO emission from biomass burning + fossil fuel + ship + ...). The resulting global files, as well as grid informations are publicly available on ICTP site :

http://clima-dods.ictp.it/regcm4/RCP_EMGLOB_PROCESSED

2. Chemical boundary conditions for important tracers are available through ICTP. We use monthly boundary condition coming from global simulations (CAM + EC-EARTH) for aerosols, following different RCP scenarios (HIST + future). For gas phase species, we now have 6 hourly chemical boundary conditions issued from the MOZART CTM. Alternatively, we can use a climatology representative of monthly average concentrations over the period 2000-2007 coming from the MOZART CTM. The oxidant fields, used in the simple sulfate scheme, is also a climatology coming from MOZART CTM. The data are available on ICTP site.

The steps to prepare the chemistry boundary conditions data are the following:

1. In case of a chemistry simulation type (CBMZ or DCCB), the global emission files must first be interpolate to the RegCM model grid using the following procedure:

- Create the RegCM model grid description file to be used by `cdo` to calculate weights for a conservative remapping interpolation:

```
$> cd $REGCM_RUN
$> ./bin/emcre_grid test_001.in
```

- Interpolate the global data on the RegCM grid with the interpolation script:

```
$> cd $REGCM_RUN
$> ./bin/interp_emissions test_001.in
```

The `cdo` program installation is mandatory in this case to perform this step. The interpolation is mass conservative and is consistent for any ratio of model resolution/global emission resolution. Note the programs and script uses the same root path of `terrain` and `icbc` programs for input and data directory. By default, we expect the global emission to be at the same level than e.g. EIN15 in your data path layout. This results in a file named `*_CHEMISS.nc` of monthly emission for the whole RCP period. You don't need to reprocess the file if you change the date of your simulation, as long as you are in the RCP temporal windows (for now, Historical from 1990-2010). Which scenario to use is controlled by the `scenario` variable in the `physicsparam` parameter namelist as above in 6.1.10

2. In function of the value of the `chemsimtype` parameter, the relevant boundary conditions will be produced on the RegCM domain by running :

```
$> cd $REGCM_RUN  
$> ./bin/chem_icbc test_001.in
```

That will result in 6 hourly chemical boundary conditions in your input directory (`*_CHBC*.nc` and/or `*_AEBC*.nc`).

Run time parameters

The other `chemparam` namelist parameters, let you control run time behaviour of the model chemistry and aerosol schemes.

1. `ichsolver` : relevant for gas-phase chemistry options, activate the chemical solver CBMZ. If different from 1, there is no chemical reactions and the tracer are only emitted, transported and removed.
2. `ichsursrc` : if set different from 1, the emission term is suppressed and only boundary conditions are generating tracer in the domain.
3. `ichdrdepo` : if set different from 1, the dry deposition and sedimentation of tracers is disabled for chemistry species.
4. `ichremlsc` : if set different from 1, rainout of chemical species is disabled.
5. `ichremcvc` : if set different from 1, washout of chemical species is disabled.
6. `ichcumtra` : if set different form 1, the convective transport of tracers is disabled.
7. `= ichdiag` : if set to 1, the writing of additional diagnostics in the chemistry output is enabled. Particularly, all the 3D tendency terms (advection, turbulence, convection, boundary condition, chemistry, removal, etc.) of the tracer equation are outputted at the frequency `ichfreq`. This is usefull for budget and sensitivity studies, as well as debugging. This potentially can generate HUGE files.
8. `idirect` : Enable aerosol feedbacks on radiation.
 - if equal to 1, only aerosol radiative forcing is calculated and outputted but there is NO aerosol radiative feedback on climate. This can be viewed as a control run option.
 - if equal to 2, there is a feedback of aerosol radiative forcing on climate fields, via perturbation of the temperature tendency. This can be view as the perturbed run option.
9. `ichdustemd` : Choice for parametrisation of dust emission size distribution:
 - if set to 1, the standard scheme s used (Alfaro et al., Zakey et al., 2006)
 - if set to 2, the revised soil granulometry + Kok et al., 2011 emission size distribution is used.
10. `rdstemfac` : Scaling factor (erodibility) for tuning DUST emission flux

Sparse notes

1. Outputs are in netCDF, so process with your favorite software.
2. The flux and tendency variables, as well as radiative forcings, are accumulated and averaged between 2 output time steps (like precipitations). The concentration, burden and aerosol optical depth are instantaneous.
3. The outputs size can be huge, especially for full chemistry and diagnostic options. In the future, we might have the choice of outputting selected variables only.

Not every possible dynamical configuration has been tested for the chemistry option, so bugs might appear: please report! CLM enables to calculate on line biogenic volatile hydrocarbon emissions, as well as chemical deposition that can be used in RegCM. There are some flags to activate when compiling CLM, we will update the documentation when fully tested.

The Tiedke and Emmanuel schemes, when activated, offer a more detailed treatment of convective transport than the simple mixing hypothesis used with other schemes. The UW planetary boundary layer option integrate directly the emission and deposition flux terms as part of the calculation of trurbulent tracer tendency.

6.1.26 debugparam namelist

This namelist is used by all RegCM programs to enable/disable some debug printout. In the current release this flag is honored only by the model itself. If you are not a developer you may find this flags useless.

```
&debugparam
debug_level = 0, ! Currently value of 2 and 3 control previous DIAG flag
dbgfrq = 24.0, ! Interval for printout if debug_level >= 3 (masscheck)
/
```

Just note that with current implementation, the output file syncing is left to the netCDF library. If You want to examine step by step the output while the model is running, set the debug_level at value 3.

6.2 The CLM options

We will now discuss from the user point of view how to use the model setup which need to be activated at configure stage.

The CLM option if activated allows the user to run a simulation using the CLM surface model instead of the default BATS1e model. We will not here go in deep in the difference between the two models, read the Reference Manual for this. The executable of the model is different in the case of the CLM, and is named regcmMPICLM. Note that in the CLM case only the MPI enabled compilation is supported (no serial), and no subgridding is possible (nsg is always 1).

Enable

At configure stage (see 3.2.1), the option is to be enabled with the right command line argument to the configure script

```
--enable-clm    Supply this option if you plan on using CLM option.
```

This will enable a preprocessing flag, and build a different model executable. Note that no modifications are needed for any other part of the model, but this triggers the building of another pre-processing program, clm2rcm.

Prepare and run

The CLM configuration requires a separate namelist in the namelist input file.

```
&clmparam
dirclm = 'input/', ! CLM path to Input data produced by clm2rcm. If
                  ! relative, It should be how to reach the Input dir
                  ! from the Run dir.
clmfrq = 12.,      ! Frequency for CLM own output write
imask = 1,         ! For CLM, Type of land surface parameterization
                  ! 1 => using DOMAIN.INFO for landmask (same as BATS)
                  ! 2 => using mksrf_navyoro file landfraction for
                  !       landmask and perform a weighted average over
                  !       ocean/land gridcells; for example:
                  ! tgb = tgb_ocean*(1-landfraction)+tgb_land*landfraction
/
```

Things you need to know here:

1. The `inpter` path defined in `terrainparam` namelist described in 6.1.4 is used also by the `clm2rcm` program. See at 4.3 how to obtain needed datasets.
2. The file `pft-physiology.c070207` should be manually copied in the `dirclm` directory before running the model.
3. The `clmfrq` is relative to the output produced by the CLM model itself, and does not control the RegCM model output. To know the CLM output file content, refer to CLM 3.5 documentation.
4. The `imask = 2` option cannot be used with the `icup_lnd` or `icup_ocn` cumulus convection schemes 2, which rely on the BATS1e landmask.

In the case of CLM run, the user needs to run, after the `terrain` program, the `clm2rcm` program, and copy the `pft-physiology.c070207` in the input directory:

```
$> cd $REGCM_RUN
$> ./bin/terrain regcm.in
$> ./bin/clm2rcm regcm.in
$> cp $REGCM_GLOBEDAT/CLM/pft-physiology.c070207 input/
```

The `clm2rcm` program interpolates global land characteristics datasets to the RegCM projected grid. The content of the `pft-physiology.c070207` file are described in the `pft-physiology.c070207.readme` file. All the other pre-processing steps are just equal to the one detailed in chapter 5. To run the CLM option in the RegCM model, just substitute the executable name:

```
$> mpirun -np 2 ./bin/regcmMPICLM regcm.in
```

Note that the CLM land model is much heavier than the BATS1e model, and computing time increases.

6.3 The CLM 4.5 options

We will now discuss from the user point of view how to use the model setup which need to be activated at configure stage.

The CLM 4.5 option if activated allows the user to run a simulation using the CLM version 4.5 surface model instead of the default BATS1e model. We will not here go in deep in the difference between the two models, read the Reference Manual for this. The executable of the model is different in the case of the CLM 4.5, and is named `regcmMPICLM45`. Note that in the CLM 4.5 case only the MPI enabled compilation is supported (no serial).

Enable

At configure stage (see 3.2.1), the option is to be enabled with the right command line argument to the configure script

```
--enable-clm45 Supply this option if you plan on using CLM45 option.
```

This will enable a preprocessing flag, and build a different model executable. Note that no modifications are needed for any other part of the model, but this triggers the building of another pre-processing program, `mksurfddata`.

Prepare and run

The CLM45 configuration requires a separate entries in the namelist input file.


```

&clm_inparm
fpftcon = 'pft-physiology.c130503.nc',
fsnowoptics = 'snicar_optics_5bnd_c090915.nc',
fsnowaging = 'snicar_drdt_bst_fit_60_c070416.nc',
hist_nhtfrq = 0,
/
&clm_soilhydrology_inparm
h2osfcflag = 1,
origflag = 0,
/
&clm_hydrology1_inparm
oldfflag = 0,
/

```

Things you need to know here:

1. The namelist 6.1.5 is used also by the `mksurfddata` program. See at 4.4 how to obtain needed datasets.
2. The `hist_nhtfrq` is relative to the output produced by the CLM 4.5 model itself, and does not control the RegCM model output.
3. To know the CLM 4.5 output file content, refer to CLM 4.5 documentation.

In the case of CLM 4.5 run, the user needs to run, after the `terrain` program, the `mksurfddata` program.

```

$> cd $REGCM_RUN
$> ./bin/terrain regcm.in
$> ./bin/mksurfddata regcm.in
$> cp $REGCM_GLOBEDAT/CLM/pft-physiology.c070207 input/

```

All the other pre-processing steps are just equal to the one detailed in chapter 5. To run the CLM 4.5 option in the RegCM model, just substitute the executable name:

```

$> mpirun -np 2 ./bin/regcmMPICLM45 regcm.in

```

6.4 Sensitivity experiments hint

Although the LBC forcing does provide a constraint for the model, as any RCM, RegCM4 is characterized by a certain level of internal variability due to its non-linear processes (e.g. convection).

For example, if small perturbations are introduced in the initial or lateral boundary conditions, the model will generally produce different patterns of, e.g. precipitation, that appear as (sometimes seemingly organized) noise when compared to the control simulation.

This noise depends on domain size and climatic regimes, for example it is especially pronounced in warm climate regimes (e.g. tropics or during the summer season) and large domains.

When doing for example sensitivity experiments to model modifications, e.g. to land use change, this internal variability “noise” can be misinterpreted as a model response to the factor modified.

Users of RegCM4 should be aware of this when they do sensitivity experiments. The best way to filter out this noise is to perform ensembles of simulations and look at the ensemble averages to extract the real model response from the noise.

Chapter 7

How To

7.1 Introduction

In this section we will describe and show how to enable/disable and control some of the physical packages of the RegCM model.

7.1.1 1D Lake Model

Chapter 8

Postprocessing tools

The new netCDF output format allows users to use a number of general purpose tools to postprocess model output files. We will in this section do a quick review of some of the Open Source and Free Software ones.

8.1 Command line tools

Three major set of tools may help you do even complex calculation just from command line prompt.

8.1.1 netCDF library tools

The netCDF library itself offers three basic tools to play with netCDF archived data.

- `ncdump` program, generates a text representation of a specified netCDF file on standard output. The text representation is in a form called CDL (“network Common Data form Language”) that can be viewed, edited, or serve as input to `ncgen`, thus `ncdump` and `ncgen` can be used as inverses to transform data representation between binary and text representations. `ncdump` may also be used as a simple browser for netCDF datasets, to display the dimension names and lengths; variable names, types, and shapes; attribute names and values; and optionally, the values of data for all variables or selected variables in a netCDF dataset. Sample usage patterns:

1. Look at the structure of the data in the netCDF dataset:

```
ncdump -c test_001_SRF.1990060100.nc
```

2. Produce a fully-annotated (one data value per line) listing of the data for the variables time and tas, using FORTRAN conventions for indices, and show the floating-point data with only four significant digits of precision and the time values with ISO format:

```
ncdump -v time,tas -p 4 -t -f \  
fortran test_001_SRF.1990060100.nc
```

- `ncgen` program, the reverse of the `ncdump` program: generates a netCDF file or a C or FORTRAN program that creates a netCDF dataset from a CDL input. Sample usage patterns:

1. From a CDL file, generate a binary netCDF file:

```
ncgen -o test_001_SRF.1990060100_modif.nc \  
test_001_SRF.1990060100.cdl
```

2. From a CDL file, generate a Fortran program to write the netCDF file:

```
ncgen -f test_001_SRF.1990060100.cdl > prog.f
```

- `nccopy` utility copies an input netCDF file to an output netCDF file, in any of the four format variants, if possible, and in function of the selected output format add compression filter and/or data chunking. Sample usage patterns:

1. Convert a netCDF dataset to a netCDF 4 classic model compressed data file using shuffling to enhance compression level:

```
nccopy -k 4 -d 9 -s test_001_SRF.1990060100.nc \
    test_001_SRF.1990060100_compressed.nc
```

You can also find, in the `Tools/Programs/RegCM_read` directory under `$REGCM_ROOT` a sample program to read an output file using the netCDF library you can modify to fit your needs.

8.1.2 NetCDF operators NCO

This set of tools can be considered a swiss army knife to manage netCDF datasets. There are multiple operators, and Each operator takes netCDF files as input, then operates (e.g., derives new data, averages, hyperslabs, manipulates metadata) and produces a netCDF output file. The single-command style of NCO allows users to manipulate and analyze files interactively, or with simple scripts that avoid some overhead of higher level programming environments. The major tools are:

- `ncap2` netCDF Arithmetic Processor
- `ncatted` netCDF Attribute Editor
- `ncbo` netCDF Binary Operator
- `ncea` netCDF Ensemble Averager
- `ncecat` netCDF Ensemble Concatenator
- `ncflint` netCDF File Interpolator
- `ncks` netCDF Kitchen Sink
- `ncpdq` netCDF Permute Dimensions Quickly, Pack Data Quietly
- `ncra` netCDF Record Averager
- `ncrcat` netCDF Record Concatenator
- `ncrename` netCDF Renamer
- `ncwa` netCDF Weighted Averager

A comprehensive user guide can be found at:

<http://nco.sourceforge.net/nco.html>

Sample usage patterns:

1. Get value of `tas` variable at a particular point for all timesteps with a prescribed format one per line on stdout:

```
ncks -C -H -s "%.2f\n" -v tas -d iy,16 -d jx,16 \
    test_001_SRF.1990060100.nc
```

2. Extract one timestep of `tas` from a file and save into a new netCDF file:

```
ncks -c -v tas -d time,6 test_001_SRF.1990060100.nc \
    test_001_SRF.1990060212.nc
```

3. Cat together a year worth of output data for the single `tas` variable into a single file:

```
ncrcat -c -v tas test_001_SRF.1990??0100.nc \
    test_001_T2M.1990.nc
```

4. Get the DJF mean value of the temperature from a multiyear run:

```
ncra -c -v tas test_001_SRF.????120100.nc \  
          test_001_SRF.????010100.nc \  
          test_001_SRF.????020100.nc \  
          test_001_DJF_T2M.nc
```

We strongly encourage you to read the on-line user guide of the NCO tools. You will for sure get a boost on your data manipulation and analysis skills.

8.1.3 Climate data Operators CDO

The monolithic `cdo` program from the Max Planck Institut für Meteorologie implements a really comprehensive collection of command line Operators to manipulate and analyse Climate and NWP model Data either in netCDF or GRIB format. There are more than 400 operators available, covering the following topics:

- File information and file operations
- Selection and Comparison
- Modification of meta data
- Arithmetic operations
- Statistical analysis
- Regression and Interpolation
- Vector and spectral Transformations
- Formatted I/O
- Climate indices

We won't make here a comprehensive analysis of this tool, but you can find some ideas in the `PostProc` directory on `$REGCM_ROOT` reading the two sample average and `regrid` scripts, which use a combination of NCO programs and `cdo` operators to reach goal. A very simple usage pattern for example to obtain a monthly mean is:

```
cdo monmean test_001_T2M.1990.nc
```

8.2 GrADS program

This tool is the one mostly used at ICTP to analyze and plot model output results. It can be used either as an interactive tool either as a batch data analysis tool. We have already written in chapter 5 about the helper program `GrADSNcPlot` which can be used to interactively plot model output results. We will here detail why an helper program is needed and how it does work. For information regarding the `grads` program itself, a comprehensive guide may be found at:

<http://www.iges.org/grads/gadoc/users.html>

8.2.1 GrADS limits

The `grads` program is powerful, yet has limits:

1. Only the equirectangular projection or Plate Carrée is supported. Some other projections can be used through a `pdef` entry in the `CTL` file using the internal direct reprojection engines, but not all RegCM supported projections are supported using direct engine.

2. NetCDF format allows multidimensional variables, while `grads` supports just four dimensional (time,level,latitude,longitude) variables.

Luckily, these limits can be exceeded, carefully telling `grads` the RegCM data structure using the CTL file and one ancillary proj file:

1. The `grads` program allows usage of the `pdef BILIN` option in the CTL file, which allows the user to specify a supplementary file name. In this file are stored three lat-lon floating-point grids which have for each point on the equirectangular grid the indexes `i,j` on the projected grid, as well as wind rotation values.
2. The `grads` program allows identifying four dimensional slices of a multidimensional variable as new variables, providing them a unique name. This is how we are able to see in `grads` chemical output variables.

While the `GrADSNcPlot` program allows interactive plotting and after quitting the `grads` program removes the CTL file and the proj file, the `GrADSNcPrepare` program only creates this two files, allowing share of the proj file between multiple CTL files sharing the same RegCM domain (i.e. it creates just only once the proj file). To use the `grads` program, you need to have both this ancillary files together with the data netCDF file.

To create the CTL file for the history CLM output file, you need to provide to the helper programs the path to both the history CLM and the RegCM DOMAIN file, as in:

```
$ GrADSNcPrepare clmoutput.clm2.h0.2000-07-30-00000.nc test_DOMAIN000.nc
```

A collection of sample `grads` scripts commonly used at ICTP to plot simulation results can be found in the `Tools/Scripts/GrADS` directory under `$REGCM_ROOT`.

8.3 CISL's NCL : NCAR Command Language

This awesome tool from NCAR is an interpreted language designed for scientific data analysis and visualization. Noah Diffenbaugh and Mark Snyder have created a website dedicated to visualizing RegCM3 output using the NCAR Command Language (NCL). These scripts were built using RegCM3 model output converted to netCDF using an external converter. They have been adapted to serve as very basic example scripts to process a native RegCM 4.2 output data file or do some data analysis using the NCL language and are available in the `Tools/Scripts/NCL/examples` directory. Travis O'Brien from the User Community also contributed sample scripts, which may be found under the `Tools/Scripts/NCL` directory.

8.4 R Statistical Computing Language

The R statistical computing language is able with an add on package to load into internal data structure a meteorological field read from a netCDF RegCM output. A sample script to load and plot the 2m Temperature at a selected timestep can be used as a reference to develop a real powerful statistical analysis of model results: it is under `Tools/Scripts/R`.

8.5 Non free tools

Note that the netCDF format, using plugins or native capabilities, allows clean access to model output from a number of non free tools like Matlab™ or IDL™.

For a more complete list of tools, you are invited to scroll down the very long list of tools at:

<http://www.unidata.ucar.edu/software/netcdf/software.html>

Chapter 9

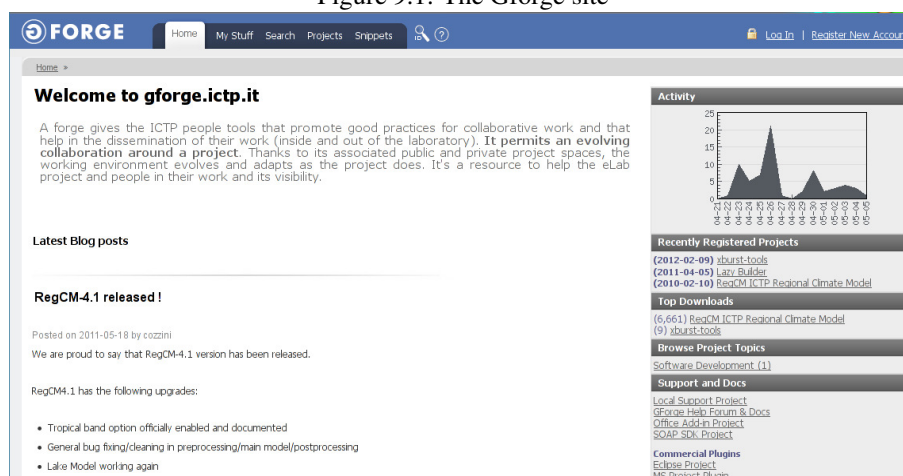
Getting help and reporting bugs

9.1 The Gforge site

A new welcoming home for the RegCM Community has been built with the help of Italian National Research Council CNR Democritos Group on the e-science Lab E-Forge web site:

<https://gforge.ictp.it/gf/project/regcm>

Figure 9.1: The Gforge site



On this site you have access with a simple registration to a friendly bug tracking system under the *tracker* link, allowing the users to post problems and bugs they discover.

It allows posting also of files to give you the opportunity to provide as much information as possible about the environment the model is running at your institution, helping us better understand and solve efficiently your problems.

Help us grow the model to fit your requirements, giving the broader user community the benefit of a valuable tool to do better research.

Chapter 10

Appendices

We will review here a sample installation session of software needed to install the RegCM model.

The starting point is here a Linux system on a multicore processor box, and the final goal is to have an optimized system to run the model. I will use `bash` as my shell and assume that GNU development tools like `make`, `sed`, `awk` are installed as part of the default Operating System environment as is the case in most Linux distro. I will require also for commodity a command line web downloader such as `curl` installed on the system, along its development libraries to be used to enable OpenDAP remote data access protocol capabilities of netCDF library. Standard file management tools such as `tar` and `gzip` and `wget` are also required. The symbol `$>` will stand for a shell prompt. I will assume that the process is performed as a normal system user, which will own all the toolchain. I will be now just the `regcm` user.

10.1 Identify Processor

First step is to identify the processor to know its capabilities:

```
$> cat /proc/cpuinfo
```

This command will ask to the operating system to print processor informations. A sample answer on my laptop is:

```
processor       : 0
vendor_id      : GenuineIntel
cpu family     : 6
model          : 30
model name     : Intel(R) Core(TM) i7 CPU           Q 740  @ 1.73GHz
stepping       : 5
cpu MHz        : 933.000
cache size     : 6144 KB
physical id    : 0
siblings       : 8
core id        : 0
cpu cores      : 4
apicid         : 0
initial apicid : 0
fpu            : yes
fpu_exception  : yes
cpuid level    : 11
wp             : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge
mca cmov pat pse36 clflush dts acpi mmx fxsr sse sse2 ss ht tm pbe syscall
```

```

nx rdtscp lm constant_tsc arch_perfmon pebs bts rep_good nopl xtopology
nonstop_tsc aperfmperf pni dtes64 monitor ds_cpl vmx smx est tm2 ssse3
cx16 xtpr pdcm sse4_1 sse4_2 popcnt lahf_lm ida dts tpr_shadow vnmi
flexpriority ept vpid
bogomips      : 3467.81
clflush size  : 64
cache_alignment : 64
address sizes  : 36 bits physical, 48 bits virtual
power management:

```

repeated eight time with Processor Ids from 0 to 7: I have a Quad Core Intel with Hyperthreading on (this multiply by 2 the reported processor list). The processor reports here also to support Intel Streaming SIMD Extensions V4.2, which can be later used to speed up code execution vectorizing floating point operation on any single CPU core.

10.2 Chose compiler

Depending on the processor, we can chose which compiler to use. On a Linux box, we have multiple choices:

- GNU Gfortran
- G95
- Intel ifort compiler
- Portland compiler
- Absoft ProFortran
- NAG Fortran Compiler

and for sure other which I may not be aware of. All of these compilers have pros and cons, so I am just for now selecting one in the pool only to continue the exposition. I am not selecting the trivial solution of Gfortran as most Linux distributions have it already packaged, and all the other required software as well (most complete distribution I am aware of for this is Fedora: all needed software is packaged and it is a matter of yum install).

So let us assume I have licensed the Intel Composer XE Professional Suite 13.0.0 on my laptop. My system administrator installed it on the default location under `/opt/intel`, and I have my shell environment update loading vendor provided script:

```
$> source /opt/intel/bin/compilervars.sh intel64
```

With some modification (the path, the script, the arguments to the script), same step is to be performed for all non-GNU compilers in the above list, and is documented in the installation manual of the compiler itself.

In case of Intel, to check the correct behaviour of the compiler, try to type the following command:

```

$> ifort --version
ifort (IFORT) 13.0.0 20120731
Copyright (C) 1985-2012 Intel Corporation. All rights reserved.

```

I am skipping here any problem that may arise from license installation for any of the compilers, so I am assuming that if the compiler is callable, it works. As this step is usually performed by a system administrator on the machine, I am assuming a skilled professional will take care of that.

10.3 Environment setup

We will now use the `prereq_install` shell script provided in the `Tools/Scripts` directory. Given the above environment, we can edit the file and decomment the lines relative to the selected compiler.

```
# Working CC Compiler
#CC=gcc
CC=icc
#CC=pgcc
# Working C++ Compiler
#CXX=g++
CXX=icpc
#CXX=pgCC
# Working Fortran Compiler
#FC=gfortran
FC=ifort
#FC=pgf90
# Destination directory
DEST=/home/regcm/intelsoft
```

I am now ready to compile software.

10.4 Pre requisite library installation

The help script will build netCDF V4 and MPI libraries to be used to compile the RegCM model.

Then we can just execute the script:

```
$> ./prereq_install.sh
This script installs the netCDF/mpi librares in the

/home/regcm/intelsoft

directory. If something goes wrong, logs are saved in

/home/regcm/intelsoft/logs
```

```
Downloading ZLIB library...
Downloading HDF5 library...
Downloading netCDF Library...
Downloading MPICH2 Library...
Compiling MPI library.
Compiled MPI library.
Compiling zlib Library.
Compiled zlib library.
Compiling HDF5 library.
Compiled HDF5 library.
Compiling netCDF Library.
Compiled netCDF C library.
Compiled netCDF Fortran library.
Done!
To link RegCM with this librares use:
```

```
PATH=/home/regcm/intelsoft/bin:$PATH ./configure \
```

```
CC=icc FC=ifort \  
CPPFLAGS=-I/home/regcm/intelsoft/include \  
LDFLAGS=-L/home/regcm/intelsoft/lib \  
LIBS="-lnetcdff -lnetcdf -lhdf5_hl -lhdf5 -lz"
```

The admins who must compile the pre requisite libraries are invited to look at the script, identifying the various steps. The normal user should be content of the last printout message which details how to use the just built libraries to compile RegCM model sources against. At run time an environment variable must be added to set correct paths:

```
$> export PATH=/home/regcm/intelsoft/bin:$PATH
```

The above is needed to be repeated for any shell which is used to run RegCM programs, and can be appended for convenience on the user shell startup scripts.

Bibliography

Giorgi, F., Regcm version 4.1 reference manual, *Tech. rep.*, ICTP Trieste, 2011.

O'Brien, T. A., L. C. Sloan, and M. A. Snyder, Can ensembles of regional climate model simulations improve results from sensitivity studies?, *Climate Dynamics*, 37, 1111–1118, doi:10.1007/s00382-010-0900-5, 2011.

R. Courant, K. F., and H. Lewy, über die partiellen differenzengleichungen der mathematischen physik, *Mathematische Annalen*, 100(1), 32–74, 1928.

Rew, R. K., and G. P. Davis, Netcdf: An interface for scientific data access, *IEEE Computer Graphics and Applications*, 10(4), 76–82, 1990.

GNU Free Documentation License
Version 1.3, 3 November 2008

Copyright (C) 2000, 2001, 2002, 2007, 2008 Free Software Foundation, Inc.
<<http://fsf.org/>>

Everyone is permitted to copy and distribute verbatim copies
of this license document, but changing it is not allowed.

0. PREAMBLE

The purpose of this License is to make a manual, textbook, or other functional and useful document "free" in the sense of freedom: to assure everyone the effective freedom to copy and redistribute it, with or without modifying it, either commercially or noncommercially. Secondly, this License preserves for the author and publisher a way to get credit for their work, while not being considered responsible for modifications made by others.

This License is a kind of "copyleft", which means that derivative works of the document must themselves be free in the same sense. It complements the GNU General Public License, which is a copyleft license designed for free software.

We have designed this License in order to use it for manuals for free software, because free software needs free documentation: a free program should come with manuals providing the same freedoms that the software does. But this License is not limited to software manuals; it can be used for any textual work, regardless of subject matter or whether it is published as a printed book. We recommend this License principally for works whose purpose is instruction or reference.

1. APPLICABILITY AND DEFINITIONS

This License applies to any manual or other work, in any medium, that contains a notice placed by the copyright holder saying it can be distributed under the terms of this License. Such a notice grants a world-wide, royalty-free license, unlimited in duration, to use that work under the conditions stated herein. The "Document", below, refers to any such manual or work. Any member of the public is a licensee, and is addressed as "you". You accept the license if you copy, modify or distribute the work in a way requiring permission under copyright law.

A "Modified Version" of the Document means any work containing the Document or a portion of it, either copied verbatim, or with modifications and/or translated into another language.

A "Secondary Section" is a named appendix or a front-matter section of the Document that deals exclusively with the relationship of the publishers or authors of the Document to the Document's overall subject (or to related matters) and contains nothing that could fall directly within that overall subject. (Thus, if the Document is in part a textbook of mathematics, a Secondary Section may not explain any mathematics.) The relationship could be a matter of historical connection with the subject or with related matters, or of legal, commercial, philosophical, ethical or political position regarding them.

The "Invariant Sections" are certain Secondary Sections whose titles are designated, as being those of Invariant Sections, in the notice that says that the Document is released under this License. If a section does not fit the above definition of Secondary then it is not allowed to be designated as Invariant. The Document may contain zero

Invariant Sections. If the Document does not identify any Invariant Sections then there are none.

The "Cover Texts" are certain short passages of text that are listed, as Front-Cover Texts or Back-Cover Texts, in the notice that says that the Document is released under this License. A Front-Cover Text may be at most 5 words, and a Back-Cover Text may be at most 25 words.

A "Transparent" copy of the Document means a machine-readable copy, represented in a format whose specification is available to the general public, that is suitable for revising the document straightforwardly with generic text editors or (for images composed of pixels) generic paint programs or (for drawings) some widely available drawing editor, and that is suitable for input to text formatters or for automatic translation to a variety of formats suitable for input to text formatters. A copy made in an otherwise Transparent file format whose markup, or absence of markup, has been arranged to thwart or discourage subsequent modification by readers is not Transparent. An image format is not Transparent if used for any substantial amount of text. A copy that is not "Transparent" is called "Opaque".

Examples of suitable formats for Transparent copies include plain ASCII without markup, Texinfo input format, LaTeX input format, SGML or XML using a publicly available DTD, and standard-conforming simple HTML, PostScript or PDF designed for human modification. Examples of transparent image formats include PNG, XCF and JPG. Opaque formats include proprietary formats that can be read and edited only by proprietary word processors, SGML or XML for which the DTD and/or processing tools are not generally available, and the machine-generated HTML, PostScript or PDF produced by some word processors for output purposes only.

The "Title Page" means, for a printed book, the title page itself, plus such following pages as are needed to hold, legibly, the material this License requires to appear in the title page. For works in formats which do not have any title page as such, "Title Page" means the text near the most prominent appearance of the work's title, preceding the beginning of the body of the text.

The "publisher" means any person or entity that distributes copies of the Document to the public.

A section "Entitled XYZ" means a named subunit of the Document whose title either is precisely XYZ or contains XYZ in parentheses following text that translates XYZ in another language. (Here XYZ stands for a specific section name mentioned below, such as "Acknowledgements", "Dedications", "Endorsements", or "History".) To "Preserve the Title" of such a section when you modify the Document means that it remains a section "Entitled XYZ" according to this definition.

The Document may include Warranty Disclaimers next to the notice which states that this License applies to the Document. These Warranty Disclaimers are considered to be included by reference in this License, but only as regards disclaiming warranties: any other implication that these Warranty Disclaimers may have is void and has no effect on the meaning of this License.

2. VERBATIM COPYING

You may copy and distribute the Document in any medium, either commercially or noncommercially, provided that this License, the copyright notices, and the license notice saying this License applies to the Document are reproduced in all copies, and that you add no other conditions whatsoever to those of this License. You may not use technical measures to obstruct or control the reading or further copying of the copies you make or distribute. However, you may accept

compensation in exchange for copies. If you distribute a large enough number of copies you must also follow the conditions in section 3.

You may also lend copies, under the same conditions stated above, and you may publicly display copies.

3. COPYING IN QUANTITY

If you publish printed copies (or copies in media that commonly have printed covers) of the Document, numbering more than 100, and the Document's license notice requires Cover Texts, you must enclose the copies in covers that carry, clearly and legibly, all these Cover Texts: Front-Cover Texts on the front cover, and Back-Cover Texts on the back cover. Both covers must also clearly and legibly identify you as the publisher of these copies. The front cover must present the full title with all words of the title equally prominent and visible. You may add other material on the covers in addition. Copying with changes limited to the covers, as long as they preserve the title of the Document and satisfy these conditions, can be treated as verbatim copying in other respects.

If the required texts for either cover are too voluminous to fit legibly, you should put the first ones listed (as many as fit reasonably) on the actual cover, and continue the rest onto adjacent pages.

If you publish or distribute Opaque copies of the Document numbering more than 100, you must either include a machine-readable Transparent copy along with each Opaque copy, or state in or with each Opaque copy a computer-network location from which the general network-using public has access to download using public-standard network protocols a complete Transparent copy of the Document, free of added material. If you use the latter option, you must take reasonably prudent steps, when you begin distribution of Opaque copies in quantity, to ensure that this Transparent copy will remain thus accessible at the stated location until at least one year after the last time you distribute an Opaque copy (directly or through your agents or retailers) of that edition to the public.

It is requested, but not required, that you contact the authors of the Document well before redistributing any large number of copies, to give them a chance to provide you with an updated version of the Document.

4. MODIFICATIONS

You may copy and distribute a Modified Version of the Document under the conditions of sections 2 and 3 above, provided that you release the Modified Version under precisely this License, with the Modified Version filling the role of the Document, thus licensing distribution and modification of the Modified Version to whoever possesses a copy of it. In addition, you must do these things in the Modified Version:

- A. Use in the Title Page (and on the covers, if any) a title distinct from that of the Document, and from those of previous versions (which should, if there were any, be listed in the History section of the Document). You may use the same title as a previous version if the original publisher of that version gives permission.
- B. List on the Title Page, as authors, one or more persons or entities responsible for authorship of the modifications in the Modified Version, together with at least five of the principal authors of the Document (all of its principal authors, if it has fewer than five), unless they release you from this requirement.
- C. State on the Title page the name of the publisher of the

Modified Version, as the publisher.

- D. Preserve all the copyright notices of the Document.
- E. Add an appropriate copyright notice for your modifications adjacent to the other copyright notices.
- F. Include, immediately after the copyright notices, a license notice giving the public permission to use the Modified Version under the terms of this License, in the form shown in the Addendum below.
- G. Preserve in that license notice the full lists of Invariant Sections and required Cover Texts given in the Document's license notice.
- H. Include an unaltered copy of this License.
- I. Preserve the section Entitled "History", Preserve its Title, and add to it an item stating at least the title, year, new authors, and publisher of the Modified Version as given on the Title Page. If there is no section Entitled "History" in the Document, create one stating the title, year, authors, and publisher of the Document as given on its Title Page, then add an item describing the Modified Version as stated in the previous sentence.
- J. Preserve the network location, if any, given in the Document for public access to a Transparent copy of the Document, and likewise the network locations given in the Document for previous versions it was based on. These may be placed in the "History" section. You may omit a network location for a work that was published at least four years before the Document itself, or if the original publisher of the version it refers to gives permission.
- K. For any section Entitled "Acknowledgements" or "Dedications", Preserve the Title of the section, and preserve in the section all the substance and tone of each of the contributor acknowledgements and/or dedications given therein.
- L. Preserve all the Invariant Sections of the Document, unaltered in their text and in their titles. Section numbers or the equivalent are not considered part of the section titles.
- M. Delete any section Entitled "Endorsements". Such a section may not be included in the Modified Version.
- N. Do not retitle any existing section to be Entitled "Endorsements" or to conflict in title with any Invariant Section.
- O. Preserve any Warranty Disclaimers.

If the Modified Version includes new front-matter sections or appendices that qualify as Secondary Sections and contain no material copied from the Document, you may at your option designate some or all of these sections as invariant. To do this, add their titles to the list of Invariant Sections in the Modified Version's license notice. These titles must be distinct from any other section titles.

You may add a section Entitled "Endorsements", provided it contains nothing but endorsements of your Modified Version by various parties--for example, statements of peer review or that the text has been approved by an organization as the authoritative definition of a standard.

You may add a passage of up to five words as a Front-Cover Text, and a passage of up to 25 words as a Back-Cover Text, to the end of the list of Cover Texts in the Modified Version. Only one passage of Front-Cover Text and one of Back-Cover Text may be added by (or through arrangements made by) any one entity. If the Document already includes a cover text for the same cover, previously added by you or by arrangement made by the same entity you are acting on behalf of, you may not add another; but you may replace the old one, on explicit permission from the previous publisher that added the old one.

The author(s) and publisher(s) of the Document do not by this License give permission to use their names for publicity for or to assert or imply endorsement of any Modified Version.

5. COMBINING DOCUMENTS

You may combine the Document with other documents released under this License, under the terms defined in section 4 above for modified versions, provided that you include in the combination all of the Invariant Sections of all of the original documents, unmodified, and list them all as Invariant Sections of your combined work in its license notice, and that you preserve all their Warranty Disclaimers.

The combined work need only contain one copy of this License, and multiple identical Invariant Sections may be replaced with a single copy. If there are multiple Invariant Sections with the same name but different contents, make the title of each such section unique by adding at the end of it, in parentheses, the name of the original author or publisher of that section if known, or else a unique number. Make the same adjustment to the section titles in the list of Invariant Sections in the license notice of the combined work.

In the combination, you must combine any sections Entitled "History" in the various original documents, forming one section Entitled "History"; likewise combine any sections Entitled "Acknowledgements", and any sections Entitled "Dedications". You must delete all sections Entitled "Endorsements".

6. COLLECTIONS OF DOCUMENTS

You may make a collection consisting of the Document and other documents released under this License, and replace the individual copies of this License in the various documents with a single copy that is included in the collection, provided that you follow the rules of this License for verbatim copying of each of the documents in all other respects.

You may extract a single document from such a collection, and distribute it individually under this License, provided you insert a copy of this License into the extracted document, and follow this License in all other respects regarding verbatim copying of that document.

7. AGGREGATION WITH INDEPENDENT WORKS

A compilation of the Document or its derivatives with other separate and independent documents or works, in or on a volume of a storage or distribution medium, is called an "aggregate" if the copyright resulting from the compilation is not used to limit the legal rights of the compilation's users beyond what the individual works permit. When the Document is included in an aggregate, this License does not apply to the other works in the aggregate which are not themselves derivative works of the Document.

If the Cover Text requirement of section 3 is applicable to these copies of the Document, then if the Document is less than one half of the entire aggregate, the Document's Cover Texts may be placed on covers that bracket the Document within the aggregate, or the electronic equivalent of covers if the Document is in electronic form. Otherwise they must appear on printed covers that bracket the whole aggregate.

8. TRANSLATION

Translation is considered a kind of modification, so you may distribute translations of the Document under the terms of section 4. Replacing Invariant Sections with translations requires special permission from their copyright holders, but you may include

translations of some or all Invariant Sections in addition to the original versions of these Invariant Sections. You may include a translation of this License, and all the license notices in the Document, and any Warranty Disclaimers, provided that you also include the original English version of this License and the original versions of those notices and disclaimers. In case of a disagreement between the translation and the original version of this License or a notice or disclaimer, the original version will prevail.

If a section in the Document is Entitled "Acknowledgements", "Dedications", or "History", the requirement (section 4) to Preserve its Title (section 1) will typically require changing the actual title.

9. TERMINATION

You may not copy, modify, sublicense, or distribute the Document except as expressly provided under this License. Any attempt otherwise to copy, modify, sublicense, or distribute it is void, and will automatically terminate your rights under this License.

However, if you cease all violation of this License, then your license from a particular copyright holder is reinstated (a) provisionally, unless and until the copyright holder explicitly and finally terminates your license, and (b) permanently, if the copyright holder fails to notify you of the violation by some reasonable means prior to 60 days after the cessation.

Moreover, your license from a particular copyright holder is reinstated permanently if the copyright holder notifies you of the violation by some reasonable means, this is the first time you have received notice of violation of this License (for any work) from that copyright holder, and you cure the violation prior to 30 days after your receipt of the notice.

Termination of your rights under this section does not terminate the licenses of parties who have received copies or rights from you under this License. If your rights have been terminated and not permanently reinstated, receipt of a copy of some or all of the same material does not give you any rights to use it.

10. FUTURE REVISIONS OF THIS LICENSE

The Free Software Foundation may publish new, revised versions of the GNU Free Documentation License from time to time. Such new versions will be similar in spirit to the present version, but may differ in detail to address new problems or concerns. See <http://www.gnu.org/copyleft/>.

Each version of the License is given a distinguishing version number. If the Document specifies that a particular numbered version of this License "or any later version" applies to it, you have the option of following the terms and conditions either of that specified version or of any later version that has been published (not as a draft) by the Free Software Foundation. If the Document does not specify a version number of this License, you may choose any version ever published (not as a draft) by the Free Software Foundation. If the Document specifies that a proxy can decide which future versions of this License can be used, that proxy's public statement of acceptance of a version permanently authorizes you to choose that version for the Document.

11. RELICENSING

"Massive Multiauthor Collaboration Site" (or "MMC Site") means any World Wide Web server that publishes copyrightable works and also provides prominent facilities for anybody to edit those works. A public wiki that anybody can edit is an example of such a server. A "Massive Multiauthor Collaboration" (or "MMC") contained in the site means any set of copyrightable works thus published on the MMC site.

"CC-BY-SA" means the Creative Commons Attribution-Share Alike 3.0 license published by Creative Commons Corporation, a not-for-profit corporation with a principal place of business in San Francisco, California, as well as future copyleft versions of that license published by that same organization.

"Incorporate" means to publish or republish a Document, in whole or in part, as part of another Document.

An MMC is "eligible for relicensing" if it is licensed under this License, and if all works that were first published under this License somewhere other than this MMC, and subsequently incorporated in whole or in part into the MMC, (1) had no cover texts or invariant sections, and (2) were thus incorporated prior to November 1, 2008.

The operator of an MMC Site may republish an MMC contained in the site under CC-BY-SA on the same site at any time before August 1, 2009, provided the MMC is eligible for relicensing.

ADDENDUM: How to use this License for your documents

To use this License in a document you have written, include a copy of the License in the document and put the following copyright and license notices just after the title page:

```
Copyright (c) YEAR YOUR NAME.
Permission is granted to copy, distribute and/or modify this document
under the terms of the GNU Free Documentation License, Version 1.3
or any later version published by the Free Software Foundation;
with no Invariant Sections, no Front-Cover Texts, and no Back-Cover Texts.
A copy of the license is included in the section entitled "GNU
Free Documentation License".
```

If you have Invariant Sections, Front-Cover Texts and Back-Cover Texts, replace the "with...Texts." line with this:

```
with the Invariant Sections being LIST THEIR TITLES, with the
Front-Cover Texts being LIST, and with the Back-Cover Texts being LIST.
```

If you have Invariant Sections without Cover Texts, or some other combination of the three, merge those two alternatives to suit the situation.

If your document contains nontrivial examples of program code, we recommend releasing these examples in parallel under your choice of free software license, such as the GNU General Public License, to permit their use in free software.